# A computable prime-corrected Riemann–von Mangoldt clock: an exact defect identity, the prime-information threshold, and the location of the zeros of the Riemann zeta function in mean square

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**Abstract**

Let $\gamma_k$ be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let $\tau_k$ be the midpoint Riemann–von Mangoldt clock, $N_0(\tau_k) = k - \frac{1}{2}$. The clock is close to the zeros in height, but the ordinal displacement $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ has mean square $\frac{1}{2\pi^2} \log\log T$ (Selberg, Korolev): the clock misplaces almost every zero by an unbounded number of positions. We prove, unconditionally: (A) an exact identity that turns sums of $g(k - \frac{1}{2} - F(\gamma_k))$ over consecutive ordinates into $dF$-integrals of $g(N - F)$, for every continuous $g$ and every $C^1$ model $F$ of the counting function $N$; (B) for the clock equation corrected by the Euler phase $S_X$ of the primes $p \le X$, the mean square of the ordinal defect at the ordinates is $\frac{1}{2\pi^2} \log \frac{\log T}{\log X} + O(1 + \sqrt{\log \frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, so that it is bounded if and only if $\log X \asymp \log T$; (C) an exact finite-height accounting of the gain of the one-step clock $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, $\mathcal{L}(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$; (D) with $X = T^{1/\sqrt{\log\log T}}$ the one-step clock, computed from $\tau_k$ and the primes alone, has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in height, against $(2 + o(1)) \log\log T/\log^2 T$ for the clock, and the Newton step iterated $\log^* T + O(1)$ times, with cutoffs depending only on $T$, locates the ordinates with a mean-square error of $O(1)$ mean spacings; the proof rests on an exact moment recursion, on Fujii's moments of $S(t+h) - S(t)$, and on a sampling inequality for maximal functions at points of bounded multiplicity; (E) $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_t |S(t)|/\log t$, and Backlund's method with the exponent profile generated from Bourgain's exponent pair gives $\limsup_k |\tau_k - \gamma_k| < 0.60831$. No zero of the Riemann zeta function is used in the construction of the corrected clocks.

*Keywords:* Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product, Newton iteration
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## 1. Introduction

### *1.1. The Riemann–von Mangoldt clock and the ordinates*

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$,

with multiplicity. Away from ordinates the Riemann–von Mangoldt formula reads [27, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

the argument of the Riemann zeta function being obtained by continuous variation along $2 \to 2+iT \to \frac{1}{2}+iT$ from $\arg\zeta(2) = 0$. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence $\tau_k$ defined by

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k-\frac{11}{8})}{W_0\Big((k-\frac{11}{8})/e\Big)} \qquad (k \geq 1), \tag{1.2}$$

where $W_0$ is the principal branch of the Lambert function; the Lambert form appears in the work of França and LeClair [8, 9]. The clock is explicit and lies, in phase, half way between consecutive Gram points. This paper studies its relation to the ordinates $\gamma_k$: in which sense the clock describes them, and how much prime information has to be added to it. The relation is measured in units of zeros by the ordinal displacement and by its prime-corrected version, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k), \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg\Big(1-p^{-1/2-it}\Big). \tag{1.3}$$

$u_k(X)$ measures how well the true ordinate satisfies the corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$; $|u_k(X)| < \frac{1}{2}$ means that the corrected count assigns to $\gamma_k$ its correct index.

*1.2. The approximation of the zeros by the clock and the primes*

The approximation studied here starts from the explicit numbers $\tau_k$ and corrects them with the Euler phase of finitely many primes and prime powers, by one or several Newton steps for the corrected equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ from $t = \tau_k$, with cutoffs that depend only on the height. Every corrected point is a closed expression in $k$, $\tau_k$ and the primes: the construction evaluates neither the Riemann zeta function nor any of its zeros. The ordinates $\gamma_k$ enter only in the statements, to measure the error. Theorems A and C are exact identities; Theorems B and D are asymptotic statements in mean square over the zeros with $T < \gamma_k \leq 2T$; Theorem E concerns the worst zero.

*1.3. What is known and only used*

The following facts are quoted with references in Section 3 and are not reproved.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \leq 2\pi \cdot 0.10076 = 0.6331\ldots$, from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [1], which improves those of Trudgian [29] and of Hasanalizade, Shen and Wong [12]. For large $T$ only, Trudgian proved $|S(T)| \leq 0.1013\log T$ with Huxley's exponent [28], which gives $\limsup_k |\delta_k| \leq 0.6365$.

- The ordinal displacement $s_k$ differs by less than 1 from minus the integer displacement of $\gamma_k$ with respect to the Gram intervals, whose moments are those of a normal variable with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae as proved by Korolev [17, 16]. Hence $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ and $\langle \delta_k^2\rangle \sim 2\log\log T/\log^2 T$ over $T < \gamma_k \leq 2T$.

- In $t$-mean, $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ is asymptotically normal (Selberg [23], Tsang [31]), and Selberg's approximation of $S(t)$ by prime sums up to a small power of $T$ holds in $L^{2k}(T,2T)$ [23, 24], in the form proved by Fujii [11, §3], together with Fujii's moments of $S(t+h)-S(t)$. These concern $t$-averages, not the ordinates.

- LeClair [18, Sect. 3] truncates the Euler product at a number of primes that tends to infinity with the height, and the truncated product is used in the equation of the individual zero with a fixed number of primes ($10^6$ for the $10^{100}$-th zero, [10, §XXI, Note added 7]). No error estimate in mean square over the zeros is given there.

*1.4. What this paper adds*

Let $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$ and let $\langle\cdot\rangle_{\mathcal{I}_T}$ be the mean over $\mathcal{I}_T$. Theorems A–E are unconditional. We have not found Theorems A–D in the literature; Theorem E is an application of Backlund's method in the form of [28, 29, 1] to exponents of [3, 14]. Absence from a search is weak evidence and does not establish priority.

**A. Exact defect identity** (Theorems 4.1 and 4.2). For every real $C^1$ model $F$ of the counting function, every continuous $g$ with primitive $\mathcal{G}$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum of $\mathcal{G}(u_k+\frac{1}{2}) - \mathcal{G}(u_k-\frac{1}{2})$ over consecutive ordinates equals $\int g(N-F)\,dF$ up to two boundary terms. For $g(x) = x^{2j}$ this gives $\sum u_k^{2j}$ ($\frac{1}{12}$ per zero when $j=1$); for $g(x) = e^{i\xi x}$ it identifies the characteristic function of the empirical law of $u_k$ with a $t$-integral. No hypothesis on the zeros and no monotonicity of $F$ is used.

**B. How much prime information the clock needs** (Theorems 5.1 and 5.12). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\frac{1}{c} + O\Big(1+\sqrt{\log(1/c)}\Big), \tag{1.4}$$

and the same holds with the cutoff $X_k = \max(2,(\tau_k/2\pi)^c)$ varying with the index. A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1}\log\log T$; primes up to $T^c$ keep it bounded; boundedness holds if and only if $\liminf \log X/\log T > 0$ (Corollary 5.4). Thus the growth of the number of primes asked for in [18] is necessary but not sufficient.

**C. Exact accounting for the computable clock** (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption; the residual of $\tau_k^X$ differs from $u_k(X)$ by an explicit transport term and by the exact nonnegative term $\chi_k = \frac{\gamma_k}{2\pi}(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k})$.

**D. A computable clock that approaches the zeros** (Theorems 7.3, 8.5 and 8.10). For $2 \le X \le T^{1/2}$ the residual $r_k(X) = \mathcal{L}(\tau_k)(\tau_k^X - \gamma_k)$ satisfies $\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{2\pi^2}\log\frac{1}{c} + O((1+\sqrt{\log(1/c)})(1+c^2\log\log T))$; the term $c^2\log\log T$ is the transport between $\tau_k$ and $\gamma_k$ (Lemma 7.2). With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ read off the clock itself,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\big(\tau_k^X - \gamma_k\big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}. \tag{1.5}$$

A second Newton step from $\tau_k^X$, with cutoff $T^{1/\sqrt{\log\log\log T}}$, has mean-square error $(\log\log\log\log T + O(\sqrt{\log\log\log\log T}))/\log^2 T$ (Theorem 8.5), and the $m$-th step, with the cutoffs (8.11) depending only on $T$, has error $(H_m + O(\sqrt{H_m}+1))/\log^2 T$ with absolute constants, where $H_m$ is an iterated logarithm of $T$ (Theorem 8.10); after $m^* = \log^* T + O(1)$ steps the error is $O(1)/\log^2 T$, that is, $O(1)$ mean spacings in mean square. The proof rests on an exact moment recursion, on a multiplicity bound for the corrected clock points of every level (Theorem 8.9, from Fujii's moments of $S(t+h)-S(t)$), and on a sampling inequality for maximal functions at points of bounded multiplicity. The second clock preserves the local order and spacing of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices (Proposition 8.7).

**E. The pointwise envelope** (Proposition 9.1 and Theorem 9.2). $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_{t\to\infty} |S(t)|/\log t$, also for the corrected clocks with $X \le (\log T)^{2-\varepsilon}$; and for every exponent profile $\mathfrak{e}$ of the Riemann zeta function on $[\frac12, 1]$,

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| \le \frac{1}{\log 2}\int_0^1 \frac{\mathfrak{e}(\sigma)\,d\sigma}{\sqrt{\sigma(2-\sigma)}}, \tag{1.6}$$

$\mathfrak{e}$ being extended to $[0, \frac12)$ by the functional equation. The profile generated from Bourgain's exponent pair by van der Corput's $A$-process gives $\limsup_k |\tau_k - \gamma_k| < 0.60831$, that is $\limsup_t |S(t)|/\log t < 0.096816$, against 0.1013 in [28] (Remark 9.3). This is an asymptotic bound and does not replace the explicit bounds of [1], valid for all $T \ge e$.

*Main finding.* The Euler phase of the primes up to a power of the height turns the clock from an approximation in height into an approximation in position. For the bare clock the error in units of zeros, $s_k = \mathcal{L}(\xi_k)\delta_k$, has mean square $\sim \frac{1}{2\pi^2}\log\log T$. With $X = T^c$ the error in units of zeros of the corrected clock equation has bounded mean square (Theorem B), so outside a proportion $\ll (1+\log(1/c))/A^2$ of the zeros it is at most $A$; and the computable clock, which uses only $\tau_k$ and the primes, keeps the gain up to the transport between $\tau_k$ and $\gamma_k$ (Theorem D). With $X = T^{1/\sqrt{\log\log T}}$,

$$\frac{\langle(\tau_k^X - \gamma_k)^2\rangle_{\mathcal{I}_T}}{\langle(\tau_k - \gamma_k)^2\rangle_{\mathcal{I}_T}} \sim \frac{\log\log\log T}{2\log\log T} \longrightarrow 0, \tag{1.7}$$

and in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$; iterating the Newton step brings it down to $O(1)$. The gain is a statement in mean square; for the worst zero, Proposition 9.1 shows that no cutoff $X \le (\log T)^{2-\varepsilon}$ changes $\limsup |\tau_k - \gamma_k|$, and the bound (1.6) comes from the exponents of the Riemann zeta function, not from the primes.

### *1.5. Degree of proof*

*Theorem*, *Proposition*, *Lemma* and *Corollary* are proved here unconditionally, from the external results of Section 3; *Question* names an open problem. No statement of this paper rests on a computation with the zeros. The only numbers that are not closed expressions are the values of the integral (9.1) for piecewise linear profiles with rational vertices, computed in closed form by the programs of Appendix Appendix A; the constants in the $O$-terms of Theorems B–D are absolute but not computed.

Table 1: Main results: degree of proof and role.

| result | status | role |
|---|---|---|
| Theorems 4.1, 4.2 (A) | unconditional | statistics of the errors at the zeros as $t$-integrals |
| Theorems 5.1, 5.12 (B) | unconditional | bounded error in units of zeros iff $\log X \asymp \log T$ |
| Theorem 6.1 (C) | unconditional, exact | finite-height balance of the gain of one correction |
| Theorems 7.3, 8.5, 8.10 (D) | unconditional | zero-free corrected points with mean-square error $O(1)$ mean spacings |
| Proposition 9.1, Theorem 9.2 (E) | unconditional, asymptotic | the worst error of the clock is the extreme size of $S(t)$ |
| Questions 10.1–10.3 | open | what remains to be proved |

### *1.6. Organization*

Section 2 lists the notation and Section 3 the external results. Sections 4–7 prove Theorems A–D, Section 8 iterates the Newton step uniformly in the number of steps, and Section 9 proves Theorem E. Section 10 lists what is open.

## 2. Notation

### *2.1. List of symbols*

Every symbol has one meaning throughout. The letter $\mu$ is not used (it is reserved for the Möbius function), $\pi$ is only the number $3.14159\ldots$, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration ($t$, $y$) are local.

| symbol | meaning |
|---|---|
| $\zeta(s)$ | the Riemann zeta function |
| $\rho_k = \beta_k + i\gamma_k$ | nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity |
| $\nu(\gamma)$ | number of zeros of the Riemann zeta function with ordinate $\gamma$ (multiplicities included) |
| $N(t)$, $N(t^{\pm})$ | number of zeros with $0 < \gamma \le t$; one-sided limits |
| $\vartheta$, $\Theta_0$, $\varpi$ | Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$ |
| $g_m$ | Gram points, $\vartheta(g_m) = m\pi$ |
| $N_0$, $\mathcal{L}$ | $N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ |
| $S$, $S^*$, $G$ | $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$ |
| $\mathcal{I}_T$, $\langle\cdot\rangle_I$ | $\{k : T < \gamma_k \le 2T\}$; mean over $k$ in a finite index set $I$ |
| $\tau_k$, $\tau_x$, $W_0$ | RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function |
| $s_k$, $\delta_k$, $\xi_k$ | $s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$ |

| symbol | meaning |
|---|---|
| $\Delta_k$ | integer displacement of $\gamma_k$ with respect to the Gram intervals (Theorem 3.4) |
| $\mathfrak{d}_k$ | $s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 1$ (Remark 3.15) |
| $\mathfrak{s}_T$ | $(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.15) |
| $\Lambda$, $a_n$ | von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$ |
| $A$, $A_X$, $\mathcal{P}$ | set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$; finite set of primes |
| $S_A$, $S_X$, $S_{\mathcal{P}}$ | $\sum_{n\in A} a_n \sin(t\log n)$; $S_{A_X}$; $S_{\{p^m : p\in\mathcal{P}\}}$ |
| $v_A$, $v_X$, $v_{\mathcal{P}}$ | $\frac{1}{2}\sum_{n\in A} a_n^2$; $v_{A_X}$; $v_{\{p^m : p\in\mathcal{P}\}}$ |
| $\sigma_X$, $M_X$ | bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.11) |
| $F$, $F_X$ | a real $C^1$ model of $N$; $F_X = N_0 + S_X$ |
| $u_k$, $u_k(X)$ | $k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$ |
| $\Psi_j$, $\kappa_j$ | sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$ |
| $X$, $c$, $C$, $T_0$ | prime cutoff; $c = \log X/\log T$; absolute constants |
| $Y$, $Y_0$, $S_Y^{\mathrm{pr}}$ | polynomial length; $Y_0 = T^{1/160}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$ |
| $\mathcal{J}$, $V$, $\mathcal{W}$, $\mathcal{W}_1$ | primes between $X$ and $Y_0$; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4) |
| $E_0$, $E_X$, $\mathcal{E}_2$ | $S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$ |
| $z_n$, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, $\eta_n$, $\eta_n'$ | coefficients and Dirichlet polynomials in Lemmas 5.9–5.10 |
| $\tau_k^X$, $\delta_k^X$, $r_k(X)$, $\chi_k$ | one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3) |
| $\mathcal{R}_n^{\gamma}(I)$, $\mathcal{R}_n^{\tau}(I)$ | prime-power responses at the zeros and at the clock points (6.1) |
| $\mathrm{Cr}_A(I)$, $D_X(I)$ | cross term (6.2); diagonal term (6.5) |
| $\lambda$ | real multiplier of an Euler phase |
| $\varsigma_n$, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$ | $\varsigma_n(k) = \sin(\tau_k\log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3) |
| $t_k^X$ | root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.7) |
| $\ell_n$ | $\log n$ |
| $\psi$, $t_1$, $t_2$ | test function and endpoints in Lemma 4.7 |
| $\mathfrak{g}$, $\mathfrak{G}$, $\omega$ | continuous test function and a primitive (Theorem 4.2); real frequency variable |
| $\mathfrak{t}_k(X)$, $y_k$ | transport term $S_X(\gamma_k) - S_X(\tau_k)$; one-step correction $S_X(\tau_k)/\mathcal{L}(\tau_k)$ |
| $\varphi$, $\mathfrak{h}$, $\mathcal{M}$, $\mathfrak{b}_T$, $\mathcal{U}_k$ | cutoff function; $\varphi S_X'$; centred Hardy–Littlewood maximal operator; $2\pi/\log(3T)$; sampling intervals (Lemma 7.2) |
| $X_k$, $\tilde{u}_k$, $X_\pm$, $V_\pm$ | variable cutoff; its defect; extreme cutoffs in a window; $\sum_{X_-<p\le X_+} p^{-1}$ (Theorem 5.12) |
| $\mathcal{Q}$, $\mathfrak{w}$ | maximal partial prime sum; smooth window (Lemma 5.11) |
| $\mathcal{I}_T^{\tau}$, $\tau_k^{\mathrm{loc}}$ | $\{k : T \le \tau_k < 2T\}$; one-step clock with cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ (Corollary 7.5) |
| $\mathfrak{a}$, $\mathrm{BW}_S$ | $\limsup_{t\to\infty}\lvert S(t)\rvert/\log t$; Bellotti–Wong bound for $\lvert S(t)\rvert$ (Theorem 3.2) |
| $\mathfrak{e}$, $(\mathsf{k},\mathsf{l})$ | exponent profile of the Riemann zeta function; exponent pair (Section 9) |
| $\sigma_\circ$, $\varrho$, $\varkappa$, $\sigma^\sharp$, $\delta^\sharp$ | centre and radius of the Jensen disc; power in Backlund's function; parameters of Backlund's trick (Theorem 3.14) |

| symbol | meaning |
|---|---|
| $x_k^{(j)}$, $e_{j,k}$, $\chi_{j,k}$, $\|\cdot\|_{r,I}$ | iterated Newton points (8.1); their errors; curvature terms; discrete norms (Section 8) |
| $H_j$, $c_j$, $y_k$, $\tilde{n}_j$ | $H_0 = \log\log T$, $H_j = 2\log(1/c_j)$ and the cutoff law (8.11); $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$; multiplicity functions (Lemma 8.3, Theorem 8.9) |
| $A_j(r)$, $\mathfrak{k}_j$, $m^*$, $C_\star$, $H^*$ | normalized moments of the errors, multiplicity constants, number of steps and constants of Theorem 8.10 |
| $n_{\mathsf{R}}(u)$, $C_1, C_2, C_3$ | near-ordinate count and constants of Lemma 8.8 and Theorem 8.9 |
| $\mathfrak{f}_\varkappa$, $\mathcal{Z}_\varkappa$, $F_\circ$ | Backlund's auxiliary function, its zero sum and the majorant (Theorem 3.14) |

*2.2. Counting functions*

Let $\vartheta(t) = \Im \log\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ for $t > 0$. For $t > 0$ not an ordinate [27, Thms. 9.3–9.4],

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta\Big(\tfrac{1}{2} + it\Big) = O(\log t), \tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2 + it$ and then horizontally to $\frac{1}{2} + it$. At an ordinate we keep the right-continuous $N$ and define $S$ by (2.1); $S^*(\gamma) = \frac{1}{2}(S(\gamma^-) + S(\gamma^+))$. Put

$$\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}, \qquad N_0(t) = \frac{1}{\pi}\Theta_0(t) + 1, \qquad \varpi(t) = \vartheta(t) - \Theta_0(t). \tag{2.2}$$

By Stirling's formula $\varpi(t) = \frac{1}{48t} + O(t^{-3})$ and $\varpi'(t) = O(t^{-2})$; in particular $|\varpi(t)|/\pi < \frac{1}{100}$ for $t \geq 14$. For all $t > 0$,

$$N(t) = N_0(t) + S(t) + \frac{\varpi(t)}{\pi}, \qquad \mathcal{L}(t) = N_0'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{2.3}$$

Since $N(2T) - N(T) = N_0(2T) - N_0(T) + O(\log T)$,

$$\#\mathcal{I}_T = T\mathcal{L}(T) + O(T). \tag{2.4}$$

*2.3. The RvM clock and the displacements*

$N_0$ is $C^\infty$ and strictly increasing on $(2\pi, \infty)$, with $N_0(2\pi) = -\frac{1}{8}$ and $N_0(t) \to \infty$. Hence for every real $x \geq 1$ the equation $N_0(t) = x - \frac{1}{2}$ has exactly one solution $\tau_x > 2\pi$. Writing $t = 2\pi y$, the equation is $y\log(y/e) = x - \frac{11}{8} \geq -\frac{3}{8}$; its root $y > 1$ is $y = (x - \frac{11}{8})/W_0((x - \frac{11}{8})/e)$, with $y = e$ if $x = \frac{11}{8}$ [5]. Thus

$$\tau_x = \frac{2\pi\,(x - \frac{11}{8})}{W_0\Big((x - \frac{11}{8})/e\Big)}, \qquad \frac{d\tau_x}{dx} = \frac{1}{\mathcal{L}(\tau_x)} = \frac{2\pi}{\log(\tau_x/2\pi)}, \tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k) = \pi(k - \frac{3}{2})$ and $\varpi = O(1/t)$, $\tau_k$ lies in phase half way between $g_{k-2}$ and $g_{k-1}$, up to $O(\tau_k^{-1}(\log\tau_k)^{-1})$.

**Definition 2.1.** For $k \geq 1$, $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ and $\delta_k = \tau_k - \gamma_k$.

Since $\gamma_1 > 14 > 2\pi$ and $\tau_k > 2\pi$, the mean value theorem applied to $s_k = N_0(\tau_k) - N_0(\gamma_k)$ gives

$$s_k = \mathcal{L}(\xi_k)\,\delta_k \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{2.6}$$

**Lemma 2.2.** *For every $k \geq 1$,*

$$N(\gamma_k^-) - N_0(\gamma_k) + \frac{1}{2} \leq s_k \leq N(\gamma_k^+) - N_0(\gamma_k) - \frac{1}{2}, \tag{2.7}$$

*and*

$$\left| s_k - \frac{\varpi(\gamma_k)}{\pi} \right| \leq \max\Big(|S(\gamma_k^-)|, |S(\gamma_k^+)|\Big) - \frac{1}{2}. \tag{2.8}$$

*Proof.* The indices $j$ with $\gamma_j = \gamma_k$ are exactly $N(\gamma_k^-) + 1, \dots, N(\gamma_k^+)$, so $N(\gamma_k^-) + 1 \leq k \leq N(\gamma_k^+)$; subtract $\frac{1}{2} + N_0(\gamma_k)$. By (2.3) at $\gamma_k^\pm$ and continuity of $N_0$ and $\varpi$, (2.7) reads $L + \frac{1}{2} \leq x \leq U - \frac{1}{2}$ with $x = s_k - \varpi(\gamma_k)/\pi$, $L = S(\gamma_k^-)$, $U = S(\gamma_k^+)$, and $U - L = \nu(\gamma_k) \geq 1$. Put $m = \max(|L|, |U|)$. Then $x \leq U - \frac{1}{2} \leq m - \frac{1}{2}$ and $x \geq L + \frac{1}{2} \geq -m + \frac{1}{2}$, which is (2.8). □

*2.4. Euler phases and the corrected clock*

For $n \geq 2$ let $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set $A$ of prime powers with $\sum_{n\in A} |a_n| < \infty$,

$$S_A(t) = \sum_{n\in A} a_n \sin(t \log n), \qquad v_A = \frac{1}{2}\sum_{n\in A} a_n^2; \tag{2.9}$$

the series converges absolutely and uniformly in $t$, so $S_A$ is continuous and bounded. For $X \geq 2$, $\sum_{n\in A_X} |a_n| \leq \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1} \frac{1}{mp^{m/2}} < \infty$ and, since $\arg(1-z) = -\Im \sum_{m\geq 1} z^m/m$ for $|z| < 1$ with the principal argument,

$$S_X(t) = -\frac{1}{\pi}\sum_{p\leq X} \arg\Big(1 - p^{-1/2-it}\Big) = -\frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1} \frac{\sin(mt\log p)}{m\,p^{m/2}} = S_{A_X}(t). \tag{2.10}$$

The sum of principal arguments is a continuous argument of the finite Euler product, not necessarily its principal argument. The series of derivatives converges uniformly, because $\sum_{p\leq X} \log p \sum_{m\geq 1} p^{-m/2} < \infty$; hence $S_X \in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)| \leq \sigma_X := \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1} \frac{1}{m\,p^{m/2}}, \qquad |S_X'(t)| \leq M_X := \frac{1}{\pi}\sum_{p\leq X} \frac{\log p}{\sqrt{p}-1}, \tag{2.11}$$

with $\sigma_X \ll \sqrt{X}/\log X$ and $M_X \ll \sqrt{X}$ by the prime number theorem.

**Definition 2.3.** For $X \geq 2$ let $F_X = N_0 + S_X$ and $u_k(X) = k - \frac{1}{2} - F_X(\gamma_k) = s_k - S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X = \tau_k - \frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)}, \qquad \delta_k^X = \tau_k^X - \gamma_k, \qquad r_k(X) = \mathcal{L}(\tau_k)\,\delta_k^X. \tag{2.12}$$

Since $F_X(\tau_k) - (k - \frac{1}{2}) = S_X(\tau_k)$, $\tau_k^X$ is the first step of $t \mapsto t - (F_X(t) - k + \frac{1}{2})/\mathcal{L}(t)$ from $t = \tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k) + S^*(\gamma_k) + \varpi(\gamma_k)/\pi = k - \frac{1}{2}$, so $u_k(X) = S^*(\gamma_k) - S_X(\gamma_k) + \varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

## 3. External results

The results of this section are known; they are used, not reproved.

**Theorem 3.1** ([27, Thms. 9.4, 9.9 and §9.12], [20]). *$S(t) = O(\log t)$; $\gamma_{k+1} - \gamma_k$ is bounded; and $S_1(t) = \int_0^t S(y)\,dy = O(\log t)$.*

**Theorem 3.2** (Bellotti–Wong [1, Thm. 1.1, Cor. 1.5 and (1.2)]). *For $T \ge e$, $\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\Big| \le \mathrm{BW}(T)$ and $|S(T)| \le \mathrm{BW}_S(T)$, where*

$$\mathrm{BW}_S(t) = 0.10076\log t + \min\{0.24460\log\log t + 7.20844,\ 1.68845\log\log t + 1.50956\}. \tag{3.1}$$

*Moreover $|S(T)| \le 2.5167$ for $0 \le T \le 30\,610\,046\,000$.*

Earlier explicit versions are due to Trudgian [29] and to Hasanalizade, Shen and Wong [12]. For $T$ sufficiently large, Trudgian [28, Thm. 2 and §5] proved $|S(T)| \le 0.1013\log T$ with Huxley's exponent $\frac{32}{205}$ for $\zeta(\frac12 + it)$. The last bound rests on Platt's rigorous isolation of the zeros up to that height [1, (1.2)].

**Theorem 3.3** (Selberg [23]; see [11, p. 225]). *For every integer $j \ge 1$, $\int_T^{2T} S(t)^{2j}\,dt = \kappa_j\, T(\log\log T)^j + O_j\Big(T(\log\log T)^{j-1/2}\Big)$.*

**Theorem 3.4** (Selberg; Korolev [17], [16, Thm. 4.2]). *Let $\Delta_k$ be the integer such that $g_{k+\Delta_k-2} < \gamma_k \le g_{k+\Delta_k-1}$ (in [16] the Gram points are $t_m = g_{m-1}$ and $t_{m-1} < \gamma_k \le t_m$ with $m = k + \Delta_k$), and let $0 < \varepsilon_0 < 10^{-3}$. For every fixed $j \ge 1$, as $n_0 \to \infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0} \le h_0 \le n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j} = \kappa_j h_0(\log\log n_0)^j\Big(1+o(1)\Big), \qquad \Big|\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j-1}\Big| \ll_j h_0(\log\log n_0)^{j-1}. \tag{3.2}$$

**Theorem 3.5** (Selberg's approximation in the form of Fujii [11, §3, pp. 232–233]). *For every integer $k \ge 1$, $\int_T^{2T}\Big(S(t) - S_{Y_k}^{\mathrm{pr}}(t)\Big)^{2k}dt \le (Ck^2)^{2k}\,T$ with $Y_k = T^{1/(80k)}$ and an absolute constant $C$. In particular $\int_T^{2T}(S - S_{Y_0}^{\mathrm{pr}})^4 \ll T$ with $Y_0 = Y_2 = T^{1/160}$.*

In [11, §3] Selberg's difference between $\pi^{-1}\arg L(\frac12 + it, \chi)$ and a prime sum is written as five explicit terms, each bounded on p. 233 in $L^{2k}(T, T+H)$ by $\ll k^2 H^{1/2k}$ under the hypotheses of Fujii's Main Theorem, which restrict $q$, $H$ and $h$ but not $k$. For the principal character (so that the $L$-function is the Riemann zeta function), $H = T$ and Fujii's parameter $a_2 = \frac14$, the prime sum runs over $p < x^3$ with $x = T^{a_2/60k}$, that is $p < T^{1/(80k)} = Y_k$; the five terms are bounded by $k^{1/2}$, $k^2$, $k^2$, $k^{1/2}$ and $k^{3/2}$ times $H^{1/2k}$, with absolute implied constants, whence the constant $(Ck^2)^{2k}$; and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi}p^{-1/2}\sin(t\log p)$. For $k = 2$ this is the form used in Section 5; $k = 4$, with $Y_4 = T^{1/320}$, and the general $k$ are used in Section 8.

**Theorem 3.6** (Fujii [11, Main Theorem, p. 225]). *For every integer $k \ge 1$ there is $C_k$ such that, for $T \ge T_0$ and $0 < h \le 1$,*

$$\int_T^{2T}\Big(S(t+h) - S(t)\Big)^{2k}dt \le C_k\,T\Big(\log(2 + h\log T)\Big)^k.$$

Fujii's statement is an asymptotic formula with main term $c_k H(2\log(2+h\log T))^k$ and relative error $O((Ak)^{3k}(\log(2+h\log T))^{-1/2})$ for $T^{1/2+a_2} \le H \le T$; the upper bound with $H = T$ follows for all $0 < h \le 1$. Only $k = 1, 2$ are used (Lemma 8.8).

**Theorem 3.7** (Montgomery–Vaughan [22, Cor. 3]). *For complex $z_n$, $\int_T^{2T} \Big|\sum_{n\le Y} z_n n^{-it}\Big|^2 dt = \sum_{n\le Y} |z_n|^2 \Big(T + O(n)\Big)$ with an absolute implied constant.*

(Replace $z_n$ by $z_n n^{-iT}$ to pass from $[0,T]$ to $[T,2T]$.)

**Theorem 3.8** (Mertens [21]; see [26, Ch. I.1, Thms. 1.8 and 1.10]). *For $y \ge 2$, $\sum_{p\le y} \frac{\log p}{p} = \log y + O(1)$ and $\sum_{p\le y} \frac{1}{p} = \log\log y + O(1)$.*

**Theorem 3.9** (Selberg [23]; Tsang [31]; see [6, (1.1)]). *For fixed real $y_1 < y_2$, $\frac{1}{T}\operatorname{meas}\{t \in [T,2T] : y_1 < S(t)/\mathfrak{s}_T \le y_2\} \to \frac{1}{\sqrt{2\pi}} \int_{y_1}^{y_2} e^{-y^2/2} dy$, where $\mathfrak{s}_T = (\frac{1}{2\pi^2}\log\log T)^{1/2}$.*

In [6] the statement is that $\log\zeta(\frac12 + it)/\sqrt{\log\log T}$ tends to a standard complex Gaussian for $t$ uniform in $[T,2T]$; its imaginary part is $\pi S(t)/\sqrt{\log\log T}$, of variance $\frac12$.

**Theorem 3.10** (Hardy–Littlewood [25, Thm. I.1], as quoted in [15, (1.2)]; Kinnunen [15, Thm. 1.4]). *Let $\mathcal{M}f(x) = \sup_{y>0} \frac{1}{2y}\int_{x-y}^{x+y} |f|$ be the centred maximal function of $f \in L^1_{\mathrm{loc}}(\mathbb{R})$. For $1 < p < \infty$ and $f \in L^p(\mathbb{R})$, $\|\mathcal{M}f\|_p \le A_p \|f\|_p$, where $A_p$ is bounded for $p \ge 2$. If $f$ and $f'$ belong to $L^p(\mathbb{R})$, $1 < p < \infty$, then $\mathcal{M}f$ has a locally absolutely continuous representative, $(\mathcal{M}f)' \in L^p(\mathbb{R})$, and $|(\mathcal{M}f)'| \le \mathcal{M}(f')$ almost everywhere.*

**Theorem 3.11** (Carleson [4], Hunt [13]; on $\mathbb{R}$ see [19]). *For $f \in L^2(\mathbb{R}) \cap L^4(\mathbb{R})$ with Fourier transform $\mathcal{F}f(\omega) = \int_{\mathbb{R}} f(t) e^{-i\omega t} dt$, let $\mathcal{C}f(x) = \sup_{\omega_1\in\mathbb{R}} \Big|\frac{1}{2\pi}\int_{-\infty}^{\omega_1} \mathcal{F}f(\omega) e^{i\omega x} d\omega\Big|$. Then $\|\mathcal{C}f\|_4 \ll \|f\|_4$.*

In [19] the statement is that $f \mapsto \sup_{\xi\in\mathbb{R}} |\mathcal{H}(e^{i\xi\cdot} f)|$, $\mathcal{H}$ the Hilbert transform, is bounded on $L^p(\mathbb{R})$ for $1 < p < \infty$; since the partial Fourier integral up to $\omega_1$ is $e^{i\omega_1 x}\cdot\frac12(\mathrm{Id} \pm i\mathcal{H})(e^{-i\omega_1\cdot} f)(x)$, the operator $\mathcal{C}f$ is at most $\frac12(|f| + \sup_\xi |\mathcal{H}(e^{i\xi\cdot} f)|)$.

**Theorem 3.12** ([14, Lemma 2.8 and (7.57)]; Bourgain [3, §5, Thm. 6]). *(a) If $(\mathsf{k},\mathsf{l})$ is an exponent pair, so is $A(\mathsf{k},\mathsf{l}) = \Big(\frac{\mathsf{k}}{2\mathsf{k}+2}, \frac12 + \frac{\mathsf{l}}{2\mathsf{k}+2}\Big)$. (b) If $(\mathsf{k},\mathsf{l})$ is an exponent pair with $\mathsf{l} - \mathsf{k} \ge \frac12$, then $\zeta(\mathsf{l} - \mathsf{k} + it) \ll |t|^{\mathsf{k}} \log|t|$ for $|t| \ge 2$. (c) For every $\varepsilon > 0$, $(\frac{13}{84} + \varepsilon, \frac{55}{84} + \varepsilon)$ is an exponent pair.*

For (b): [14, (7.57)] gives $\zeta(\sigma + it) \ll |t|^{(\mathsf{k}+\mathsf{l}-\sigma)/2}\log|t|$ for $\frac12 \le \sigma \le \mathsf{l} - \mathsf{k}$; take $\sigma = \mathsf{l} - \mathsf{k}$.

**Theorem 3.13** (Phragmén–Lindelöf principle in a strip [1, Lemma 2.5], [29, Lemma 3]). *Let $\sigma_1 < \sigma_2$ and $e_1 \ge e_2 \ge 0$. If $|\zeta(\sigma_i + it)| \le C_i |t|^{e_i}\log|t|$ for $|t| \ge 2$ and $i = 1, 2$, then $|\zeta(\sigma + it)| \ll |t|^{e(\sigma)}\log|t|$ uniformly for $\sigma_1 \le \sigma \le \sigma_2$ and $|t| \ge 2$, where $e$ is the linear function with $e(\sigma_i) = e_i$.*

(Apply [1, Lemma 2.5] to the entire function $(s-1)\zeta(s)$ of order 1, with exponents $1 + e_i$ and one power of the logarithm on the two lines.)

**Theorem 3.14** (Backlund's method [1, Lemmas 2.1, 2.3 and 2.4], after [29, 12])**.** *Let $T$ be large, $\varkappa \in \mathbb{N}$, $\mathfrak{f}_\varkappa(s) = \frac{1}{2}\Big[((s+iT-1)\zeta(s+iT))^\varkappa + ((s-iT-1)\zeta(s-iT))^\varkappa\Big]$, and for a disc $D(\sigma_\circ,\varrho)$ let $\mathcal{Z}_\varkappa(\sigma_\circ,\varrho) = \frac{1}{\varkappa}\sum_z \log\frac{\varrho}{|z-\sigma_\circ|}$ over the zeros $z$ of $\mathfrak{f}_\varkappa$ in $D(\sigma_\circ,\varrho)$. Let $1<\sigma_\circ<\varrho$, $\sigma^\sharp = \sigma_\circ + (\sigma_\circ-\frac12)^2/\varrho$, $\delta^\sharp = 2\sigma_\circ - \sigma^\sharp - \frac12$, and assume $0<\delta^\sharp<\frac12$.*

*(a) If $F_\circ : [-\pi,\pi]\to\mathbb{R}$ is even and $F_\circ(\theta) \ge \frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ+\varrho e^{i\theta})|$ for all $\varkappa$ and $\theta$, then there are $\varkappa_m\to\infty$ with $\limsup_m \mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho) \le \log\frac{\zeta(\sigma_\circ)}{\sqrt{(\sigma_\circ-1)^2+T^2}\,\zeta(2\sigma_\circ)} + \frac{1}{\pi}\int_0^\pi F_\circ(\theta)\,d\theta$.*

*(b) For every $\varkappa$, $\Big|\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg\Big((\sigma+iT-1)\zeta(\sigma+iT)\Big)\Big| \le \frac{\pi\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)}{2\log(\varrho/(\sigma_\circ-\frac12))} + \frac{\pi}{100} + \frac{3\pi}{2\varkappa} + \frac{\pi}{4}$.*

In (b) the term $\frac{\pi}{100}$ bounds $\frac12 E(T,\delta^\sharp)$ of [1, Lemma 2.3], by the monotonicity in $\delta^\sharp$ and the bound for $E(T,\frac12)$ in [1, Lemma 2.4]; (a) is [1, Lemma 2.1], which also requires $\sigma_\circ-\varrho<\frac12<1<\sigma_\circ<\sigma^\sharp<\sigma_\circ+\varrho$, a consequence of $1<\sigma_\circ<\varrho$.

*Remark* 3.15 (Known consequences for the clock). (a) *Moments and law of $s_k$.* Since $\vartheta(\gamma_k)/\pi \in (k+\Delta_k-2, k+\Delta_k-1]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$, one has $s_k+\Delta_k = k+\Delta_k-\frac12-N_0(\gamma_k) \in [-\frac12+\frac{\varpi(\gamma_k)}{\pi}, \frac12+\frac{\varpi(\gamma_k)}{\pi})$, so $s_k = -\Delta_k+\mathfrak{d}_k$ with $|\mathfrak{d}_k|<1$. The index set $\mathcal{I}_T$ is the integer interval $(N(T), N(2T)]$, of length $\asymp T\log T$. Cover it by consecutive windows $(n_0, n_0+h_0]$ with $h_0 = \lfloor n_0^{27/82+0.95\varepsilon_0}\rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0 = O(T^{-1/2}\#\mathcal{I}_T)$ indices; in each window $\log\log n_0 = \log\log T + o(1)$. Summing (3.2) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle\Delta_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j(1+o(1))$ and $\langle\Delta_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j((\log\log T)^{j-1})$. Expanding $s_k^r = (-\Delta_k+\mathfrak{d}_k)^r$ and bounding $\langle|\Delta_k|^{r-i}\rangle_{\mathcal{I}_T} \le \langle\Delta_k^{2\lceil r/2\rceil}\rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2\rceil} \ll (\log\log T)^{(r-i)/2}$ for $1\le i\le r$, one obtains for every fixed $j\ge1$

$$\langle s_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j\Big(1+o(1)\Big), \qquad \langle s_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j\Big((\log\log T)^{j-1}\Big). \tag{3.3}$$

With $\mathfrak{s}_T^2 = \frac{\log\log T}{2\pi^2}$, so that $\kappa_j(\log\log T)^j = \frac{(2j)!}{j!2^j}\mathfrak{s}_T^{2j}$, the normalized moments $\langle(s_k/\mathfrak{s}_T)^r\rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable $Z$ for every $r\ge1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1} \ll_j (\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.2) that gives the sign. The normal law is determined by its moments, so the method of moments [2, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of $Z$. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2\rangle_{\mathcal{I}_T}\sim\frac{1}{2\pi^2}\log\log T$, $\limsup_k|s_k|=\infty$, and $\#\{k\in\mathcal{I}_T : |s_k|\le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0>0$; for these the even moments suffice.

(b) *Bound for $\delta_k$.* Theorem 3.2, applied as $T\to\gamma_k^\pm$ (both sides have one-sided limits), gives $|N(\gamma_k^\pm) - N_0(\gamma_k) + \frac78| \le \mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k) - \frac38 \le s_k \le \mathrm{BW}(\gamma_k) - \frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi,\infty)$ and $s_k = \int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k)+3/8}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{\mathrm{BW}(\gamma_k)-11/8}{\mathcal{L}(\gamma_k)} \qquad (k\ge1). \tag{3.4}$$

Hence $\delta_k = O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k)\to1$, and $\limsup_k|\delta_k| \le 2\pi\cdot0.10076 = 0.6331\ldots$, an immediate consequence of [1].

## 4. Theorem A: an exact defect identity at the zeros

**Theorem 4.1.** *Let $1 \le k_1 < k_2$ be indices, $F$ a real $C^1$ function on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2j} F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac{1}{2}+u_{k_1})^{2j+1} + (\frac{1}{2}-u_{k_2})^{2j+1}}{2j+1}, \tag{4.1}$$

*where*

$$\Psi_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{l=0}^{j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l} = u^{2j} + \sum_{l<j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1} u^{2l}. \tag{4.2}$$

*In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,*

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2} F'(t)\,dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2}+u_{k_1})^3 + (\frac{1}{2}-u_{k_2})^3}{3}. \tag{4.3}$$

*Proof. Existence.* On $[\gamma_{k_1}, \gamma_{k_2}]$ the function $N$ is a step function with finitely many jumps (Theorem 3.1) and $F$, $F'$ are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

*One gap.* Fix $k$ with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly $k$ ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is $C^1$ with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of $F$ is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\,dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2}+u_k)^{2j+1} + (\frac{1}{2}-u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2}+u_k)^{2j+1} + (-\frac{1}{2}-u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every $k$, and the values of $N$ at the finitely many ordinates do not affect the integrals.

*Summation and boundary terms.* Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each $k$ with $k_1 < k < k_2$ contributes $(\frac{1}{2}+u_k)^{2j+1}$ from the gap starting at $\gamma_k$ and $(\frac{1}{2}-u_k)^{2j+1}$ from the gap ending at $\gamma_k$, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2}+u_{k_1})^{2j+1}$ and $(\frac{1}{2}-u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}$ the odd powers of $u$ cancel and $u^{2l}$ has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. □

**Theorem 4.2.** *Let $k_1 < k_2$, $F$ and $u_k$ be as in Theorem 4.1, and let $\mathfrak{g} : \mathbb{R} \to \mathbb{C}$ be continuous with a primitive $\mathfrak{G}$. Then*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \mathfrak{g}\Big(N(t) - F(t)\Big) F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1} \Big[\mathfrak{G}\Big(u_k + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_k - \tfrac{1}{2}\Big)\Big] + \mathfrak{G}\Big(u_{k_1} + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_{k_2} - \tfrac{1}{2}\Big). \tag{4.5}$$

*Proof.* The integrand is bounded and piecewise continuous, as in the proof of Theorem 4.1. If $\gamma_k < \gamma_{k+1}$, then $N = k$ on $(\gamma_k, \gamma_{k+1})$ and $t \mapsto -\mathfrak{G}(k - F(t))$ is $C^1$ with derivative $\mathfrak{g}(k - F(t))F'(t)$, so the integral over $[\gamma_k, \gamma_{k+1}]$ is $\mathfrak{G}(k - F(\gamma_k)) - \mathfrak{G}(k - F(\gamma_{k+1})) = \mathfrak{G}(u_k + \frac{1}{2}) - \mathfrak{G}(u_{k+1} - \frac{1}{2})$. If $\gamma_k = \gamma_{k+1}$, the integral is 0 and $u_{k+1} = u_k + 1$, so the same formula holds. Summing over $k_1 \le k < k_2$, each $k$ with $k_1 < k < k_2$ contributes $\mathfrak{G}(u_k + \frac{1}{2})$ from the gap that starts at $\gamma_k$ and $-\mathfrak{G}(u_k - \frac{1}{2})$ from the gap that ends there. □

For $\mathfrak{g}(x) = x^{2j}$ this is (4.1), since $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = \Psi_j(u)$.

**Corollary 4.3.** *For real $\omega \neq 0$,*

$$\frac{2\sin(\omega/2)}{\omega} \sum_{k=k_1+1}^{k_2-1} e^{i\omega u_k} = \int_{\gamma_{k_1}}^{\gamma_{k_2}} e^{i\omega(N(t)-F(t))} F'(t)\, dt - \frac{e^{i\omega(u_{k_1}+1/2)} - e^{i\omega(u_{k_2}-1/2)}}{i\omega}. \tag{4.6}$$

*Proof.* Theorem 4.2 with $\mathfrak{g}(x) = e^{i\omega x}$ and $\mathfrak{G}(x) = e^{i\omega x}/(i\omega)$, for which $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = e^{i\omega u} \cdot 2\sin(\omega/2)/\omega$. □

**Corollary 4.4** (Signed law of $s_k$ from the $t$-law of $S$)**.** *For fixed real $\eta$, $\langle e^{i\eta s_k/\mathfrak{s}_T}\rangle_{\mathcal{I}_T} \to e^{-\eta^2/2}$. Hence the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to the standard normal law.*

*Proof.* Let $\eta \neq 0$, $\omega = \eta/\mathfrak{s}_T$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that $\{k_1+1, \dots, k_2-1\} = \mathcal{I}_T$. Apply (4.6) with $F = N_0$, so $u_k = s_k$ and $N - F = G$. By Theorem 3.1 the intervals $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$ have bounded length, so the integral equals $\int_T^{2T} e^{i\omega G}\mathcal{L}\, dt + O(\log T)$; the boundary term is $O(1/|\omega|) = O(\mathfrak{s}_T)$; and $2\sin(\omega/2)/\omega = 1 + O(\omega^2)$. Divide by $\#\mathcal{I}_T = \int_T^{2T} \mathcal{L}\, dt + O(\log T)$. On $[T, 2T]$, $|e^{i\omega G} - e^{i\omega S}| \le |\omega||\varpi|/\pi \ll (\mathfrak{s}_T T)^{-1}$ and $\mathcal{L}(t) = \mathcal{L}(T)(1 + O(1/\log T))$, whence

$$\langle e^{i\omega s_k}\rangle_{\mathcal{I}_T} = \frac{1}{T}\int_T^{2T} e^{i\eta S(t)/\mathfrak{s}_T} dt + O\Big(\omega^2 + \frac{1}{\log T}\Big).$$

By Theorem 3.9 and the definition of convergence in distribution, applied to the bounded continuous function $y \mapsto e^{i\eta y}$, the integral tends to $e^{-\eta^2/2}$, and $\omega \to 0$. The last statement is Lévy's continuity theorem [2, Thm. 26.3]. □

*Remark* 4.5. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j} dx$ is the mean of $x^{2j}$ over the unit interval centred at $u$; in particular $\Psi_1(u) = u^2 + \frac{1}{12}$ and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2 dx$. The constant comes from the unit jump of $N - F$ at each zero, through the primitive in Theorem 4.2; when $F$ is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identities remain exact with the integral counted with sign. With $F = N_0$, Theorem 3.3 and induction on $j$ through the lower-degree terms of $\Psi_j$ recover the even moments in (3.3) without Gram intervals; Corollary 4.4 gives the sign without Korolev's odd moments. The main purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

**Corollary 4.6.** *For $X \ge 2$ and $1 \le k_1 < k_2$,*

$$\begin{aligned}\sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = {} & \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(S - S_X + \frac{\varpi}{\pi}\Big)^2 \big(\mathcal{L} + S_X'\big) dt \\ & - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}.\end{aligned} \tag{4.7}$$

*Proof.* $F_X$ is $C^\infty$ on $(0,\infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

**Lemma 4.7** (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1<\gamma_k\le t_2} s_k\psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\, G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.8}$$

*Proof.* On $[t_1, t_2]$ there are finitely many ordinates, none at $t_1$ or $t_2$. The function $G = N - N_0$ is right-continuous and of bounded variation, $C^\infty$ with $G' = -\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate $\gamma$. Let $G^*(\gamma) = \frac{1}{2}(G(\gamma^-) + G(\gamma^+))$. The indices with $\gamma_k = \gamma$ are $N(\gamma^-) + 1, \dots, N(\gamma^-) + \nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k - \frac{1}{2}) = \nu(\gamma)N(\gamma^-) + \frac{\nu(\gamma)^2}{2} = \nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, $N_0$ being continuous, $\sum_{\gamma_k=\gamma} s_k = \nu(\gamma)G^*(\gamma)$. The left side of (4.8) therefore equals the finite sum $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1, t_2]$,

$$d(G^2) = -2G\mathcal{L}\,dt + \sum_{t_1<\gamma<t_2} 2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.9}$$

because $(G^2)' = 2GG' = -2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2 - G(\gamma^-)^2 = (G(\gamma^+) + G(\gamma^-))\nu(\gamma) = 2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function $\psi$ against (4.9) gives $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma) = \frac{1}{2}\int_{(t_1,t_2]} \psi\, d(G^2) + \int_{t_1}^{t_2} G\psi\mathcal{L}\,dt$. Since $\psi$ is $C^1$ and $G^2$ is right-continuous of bounded variation and continuous at $t_1$ and $t_2$, integration by parts for Lebesgue–Stieltjes integrals [7, §3.5] gives $\int_{(t_1,t_2]} \psi\, d(G^2) = [\psi G^2]_{t_1}^{t_2} - \int_{t_1}^{t_2} G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

## 5. Theorem B: how much prime information the clock needs

**Theorem 5.1.** *There are absolute constants $T_0$ and $C$ such that for $T \ge T_0$ and every $X$ with $2 \le X \le T^{1/2}$, writing $c = \log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \le C\Big(1 + \sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c) = \log\log T - \log\log X \in [\log 2, \log\log T - \log\log 2]$. For fixed $c$ both the main term and the error are bounded; the theorem does not identify a limit at fixed $c$. It concerns the defect at the true ordinate and does not select a root of $F_X(t) = k - \frac{1}{2}$.

**Corollary 5.2** (fixed cutoff)**.** *For fixed $X \ge 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\log T + O_X(\sqrt{\log\log T})$, and $\#\{k \in \mathcal{I}_T : |u_k(X)| \le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$.*

*Proof.* The first statement is (5.1). Since $|u_k(X) - s_k| \le \sigma_X$, $|u_k(X)| \le y_0$ implies $|s_k| \le y_0 + \sigma_X$; apply Remark 3.15(a). □

**Corollary 5.3** (power cutoff)**.** *For fixed $0 < c \le \frac{1}{2}$ and $X = T^c$,*

$$\limsup_{T\to\infty} \frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \le \frac{1}{2\pi^2}\log\frac{1}{c} + C\Big(1 + \sqrt{\log(1/c)}\Big), \tag{5.2}$$

*and $\#\{k \in \mathcal{I}_T : |u_k(T^c)| > w_T\} = o(\#\mathcal{I}_T)$ whenever $w_T \to \infty$. The same holds for $X = N(T)^c$ with fixed $0 < c < \frac{1}{2}$, since then $X = T^{c'}$ with $c' = c + O(\log\log T/\log T)$.*

*Proof.* (5.1) and Chebyshev's inequality. □

**Corollary 5.4** (dichotomy)**.** *Let $X = X(T)$ with $2 \le X \le T^{1/2}$. The mean square of $u_k(X)$ over $\mathcal{I}_T$ is bounded as $T \to \infty$ if and only if $\liminf_{T\to\infty} \log X/\log T > 0$. For fixed $X_0 \ge 2$ and $c \in (0, \frac12]$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(u_k(X_0)^2 - u_k(T^c)^2\Big) = \frac{1}{2\pi^2}\log\log T + O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

*Proof.* Both directions follow from (5.1), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

**Proposition 5.5** (Rounding the corrected count misassigns a positive proportion)**.** *There are absolute constants $c_* > 0$, $\kappa_0 > 0$ and $T_0$ such that for $T \ge T_0$ and $2 \le X \le T^{c_*}$,*

$$\frac{1}{T}\operatorname{meas}\Big\{t \in [T, 2T] : \Big\lfloor F_X(t) + \tfrac12\Big\rfloor \ne N(t)\Big\} \ge \kappa_0.$$

*Proof.* If $\lfloor F_X(t) + \frac12\rfloor = N(t)$ then $|N(t) - F_X(t)| \le \frac12$, i.e. $|E| \le \frac12$ with $E = E_X + \varpi/\pi$. Let $\mathfrak{m}$ be the measure of $\{t \in [T, 2T] : |E| > \frac12\}$. Then $\int_T^{2T} E^2 \le \frac{T}{4} + (\int_T^{2T} E^4)^{1/2}\mathfrak{m}^{1/2}$. By (5.10) and (5.11) (with $|\varpi| \ll 1/T$), $\int_T^{2T} E^2 \ge \frac{T}{2\pi^2}V - CT(1+\sqrt{V})$ and $\int_T^{2T} E^4 \le CT(1+V)^2$, so $\mathfrak{m}/T \ge (\frac{V}{2\pi^2} - C(1+\sqrt{V}) - \frac14)_+^2/(C(1+V)^2)$, which is $\ge \kappa_0 > 0$ once $V \ge V_0$, that is $c \le c_*$ by Lemma 5.8. □

Rounding the corrected count to the nearest integer therefore does not assign the correct index to the zeros, for small cutoffs, on a set of positive density: the obstruction is the size of $S - S_X$, not the sawtooth $N - N_0$. For $c$ near $\frac12$ the question depends on the constant in (5.1) (Question 10.1), and for $c > \frac12$ nothing is proved.

*Remark* 5.6. Since $\log\#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes, as in the computations reported in [10, §XXI, Note added 7], leaves almost all zeros violating the corrected equation by more than any fixed amount; the requirement that the number of primes tend to infinity [18, Sect. 3] is necessary but not sufficient. The range $c > \frac12$ is not covered. In Theorem 5.1 the cutoff is constant on each window $\mathcal{I}_T$; Theorem 5.12 below allows the cutoff $X_k = (\tau_k/2\pi)^c$, which varies at every ordinate.

*Remark* 5.7 (Localization of the ordinates). If $M_X \le \frac12\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac12\mathcal{L}(T)$ on $[T, 2T]$, so $F_X$ is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac12 \le F_X(2T)$ the equation $F_X(t) = k - \frac12$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$, and in that range the localization gains nothing in order (Proposition 7.6). For $X = T^c$, $F_X$ need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root; Theorem 7.3 replaces the root by one Newton step.

*5.1. Lemmas on prime polynomials*

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge 3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.10); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in $(X, Y_0)$ if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi} \sum_{p \in \mathcal{J}} \frac{\sin(t \log p)}{\sqrt{p}}, \qquad V = \sum_{p \in \mathcal{J}} \frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm \mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge 3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi} p^{-1/2} \sin(t \log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

**Lemma 5.8.** $V = \log(1/c) + O(1)$ *uniformly for* $\log 2 / \log T \le c \le \frac{1}{2}$.

*Proof.* By Theorem 3.8, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log \frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log \frac{1}{c} - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac{1}{2}$ both $V$ and $\log \frac{1}{c}$ are $O(1)$. □

**Lemma 5.9.** *For real* $z_n$, $2 \le n \le Y$,

$$\int_T^{2T} \Big( \sum_{n \le Y} z_n \sin(t \log n) \Big)^2 dt = \frac{T}{2} \sum_{n \le Y} z_n^2 + O\Big( \sum_{n \le Y} n z_n^2 + \Big( \sum_{n \le Y} |z_n| \Big)^2 \Big), \tag{5.5}$$

*and the same with* cos *in place of* sin.

*Proof.* With the finite sum $\mathcal{D}(t) = \sum z_n n^{-it}$, $\sum z_n \sin(t \log n) = -\Im \mathcal{D}$ and $(\Im \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 - \Re \mathcal{D}^2)$. Theorem 3.7 evaluates $\int |\mathcal{D}|^2$, and $|\int_T^{2T} (n_1 n_2)^{-it} dt| \le 2/\log(n_1 n_2) \le 2/\log 4$ gives $|\int \mathcal{D}^2| \le (2/\log 4)(\sum |z_n|)^2$. For cos use $\Re \mathcal{D}$ and $(\Re \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 + \Re \mathcal{D}^2)$. □

**Lemma 5.10.** *With absolute implied constants,*

$$\int_T^{2T} \mathcal{W}^2 \, dt = \frac{T}{2\pi^2} V + O\Big( T(1 + \sqrt{V}) \Big), \tag{5.6}$$

$$\int_T^{2T} \mathcal{W}^4 \, dt \ll T(1+V)^2, \tag{5.7}$$

$$\int_T^{2T} S_X'(t)^2 \, dt \ll T(1 + \log X)^2. \tag{5.8}$$

*Proof.* All primes involved are $\le \max(X, Y_0) \le T^{1/2}$. Lemma 5.9 with $z_p = \frac{1}{\pi\sqrt{p}}$ ($p \in \mathcal{J}$) gives $\int \mathcal{W}_1^2 = \frac{T}{2\pi^2} V + O(T^{1/2} + (\sum_{p \le T^{1/2}} p^{-1/2})^2) = \frac{T}{2\pi^2} V + O(T^{1/2})$. For $S_X^{(2)}(t) = -\frac{1}{2\pi} \sum_{p \le X} p^{-1} \sin(t \log p^2)$ the same lemma gives $\int (S_X^{(2)})^2 \ll T \sum_p p^{-2} + \#\{p \le X\} + (\sum_{p \le X} p^{-1})^2 \ll T$. Finally $|S_X^{(\ge 3)}| \le \frac{1}{\pi} \sum_p \sum_{m \ge 3} \frac{1}{m p^{m/2}} \le \frac{1}{\pi} \sum_p \frac{p^{-3/2}}{1 - p^{-1/2}} < \infty$, uniformly in $X$. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1 + y_2 + y_3)^4 \le 27(y_1^4 + y_2^4 + y_3^4)$. Let $\mathcal{D}_1(t) = \sum_{p \in \mathcal{J}} p^{-1/2} p^{-it}$, so that $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$, $\eta_{p_1 p_2} = 2(p_1 p_2)^{-1/2}$ for $p_1 < p_2$, all $n \le T$. Theorem 3.7 gives $\int |\mathcal{D}_1|^4 = \sum |\eta_n|^2 (T + O(n)) \ll T \sum |\eta_n|^2 \le T(1 + 2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t) = \sum_{p \le X} p^{-1} (p^2)^{-it}$, so that $|S_X^{(2)}| \le |\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2 = \sum_n \eta_n' n^{-it}$ over $n = (p_1 p_2)^2$

with $|\eta_n'| \le 2(p_1p_2)^{-1}$; then $\sum |\eta_n'|^2 \ll 1$ and $\sum n|\eta_n'|^2 \le 4\#\{p \le X\}^2 \le 4X^2 \le 4T$. $S_X^{(\ge 3)}$ is bounded.

For (5.8), $S_X'(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m = 1$ have $|z_p| = \frac{\log p}{\pi\sqrt{p}}$; Lemma 5.9 and Theorem 3.8 give $\ll T\log X \sum_{p\le X} \frac{\log p}{p} + \sum_{p\le X} \log^2 p + (\sum_{p\le X} \frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log X$, and $X \le T^{1/2}$. The terms $m = 2$ have $|z_{p^2}| = \frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \log p \sum_{m\ge 3} p^{-m/2} < \infty$. Combine with $(y_1+y_2+y_3)^2 \le 3(y_1^2+y_2^2+y_3^2)$. □

*5.2. Proof of Theorem 5.1*

*Proof. Integrability.* On $[T, 2T]$, $S$ is bounded (Theorem 3.1) and piecewise continuous, and $S_X$, $S_X'$, $\varpi$ are continuous; all integrals below exist.

*Second and fourth moments of $E_X$.* Let $E_0 = S - S_{Y_0}^{\mathrm{pr}}$ and $E_X = S - S_X = E_0 + \mathcal{W}$. Theorem 3.5 and Hölder's inequality give

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{5.9}$$

Let $\mathcal{E}_2 = \int_T^{2T} E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2 = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$,

$$\int_T^{2T} E_X^4\,dt \ll T(1+V)^2. \tag{5.11}$$

*Transfer to the ordinates.* Let $k_1, k_2$ be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \ne \emptyset$ for $T \ge T_0$) and apply (4.7). By Lemma 2.2 and (2.11), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S_X'|) \ll T^{1/2}\cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by $E_X^2$ costs $\ll T^{-1}\int_T^{2T}(|E_X|+1)(\mathcal{L}+|S_X'|)\,dt \ll (1+V)^{1/2}\log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2\mathcal{L}\,dt + \int_T^{2T} E_X^2 S_X'\,dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

*Main term and derivative term.* Since $0 \le \mathcal{L}(t) - \mathcal{L}(T) \le \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\Big|\int_T^{2T} E_X^2 S_X'\,dt\Big| \le \Big(\int_T^{2T} E_X^4\Big)^{1/2}\Big(\int_T^{2T} S_X'^2\Big)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\,c\Big), \tag{5.14}$$

because $\log X = c\log T$. By Lemma 5.8, $(1+V)c \ll c(1+\log\frac{1}{c}) \ll 1$ on $(0, \frac{1}{2}]$; $V \ll \log\log T$ since $c \ge \log 2/\log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.8 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2}\log\frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

*5.3. A cutoff varying with the index*

Let $\mathcal{I}_T^\tau = \{k : T \le \tau_k < 2T\}$; by (2.5), $\#\mathcal{I}_T^\tau = N_0(2T) - N_0(T) + O(1) \asymp T\log T$.

**Lemma 5.11.** *Let $T \ge T_0$ and $2 \le X_- < X_+ \le (T/3)^{1/2}$ with $X_+ \le 4X_-$; let $V_\pm = \sum_{X_-<p\le X_+} p^{-1}$ and*

$$\mathcal{Q}(t) = \max_{X_-\le y\le X_+}\Big|\sum_{X_-<p\le y} p^{-1/2-it}\Big|.$$

*Then $\mathcal{Q}$ is Lipschitz, $|\mathcal{Q}'(t)| \le \max_{X_-\le y\le X_+}|\sum_{X_-<p\le y}\log p\, p^{-1/2-it}|$ for almost every $t$, and*

$$\int_{T-3}^{2T+3}\mathcal{Q}^4\,dt \ll TV_\pm^2, \qquad \int_{T-3}^{2T+3}|\mathcal{Q}'|^4\,dt \ll T(\log X_+)^4V_\pm^2, \tag{5.15}$$

$$\langle\mathcal{Q}(\gamma_k)^2\rangle_{\mathcal{I}_T} \ll V_\pm\Big(1+\frac{\log X_+}{\log T}\sqrt{\log\log T}\Big)+\frac{X_+}{T}, \qquad \langle\mathcal{Q}(\tau_k)^2\rangle_{\mathcal{I}_T^\tau} \ll V_\pm. \tag{5.16}$$

*Proof.* If $(X_-, X_+]$ contains no prime, $\mathcal{Q} = 0$. Otherwise let $p_1 < \cdots < p_J$ be its primes and $P_j(t) = \sum_{i\le j} p_i^{-1/2-it}$ ($P_0 = 0$), so that $\mathcal{Q} = \max_{0\le j\le J}|P_j|$, a maximum of finitely many Lipschitz functions; hence $\mathcal{Q}$ is Lipschitz and differentiable almost everywhere. At a point $t$ where $\mathcal{Q}$ and all $|P_j|$ are differentiable and $\mathcal{Q}(t) = |P_j(t)|$, the function $\mathcal{Q} - |P_j| \ge 0$ has a minimum, so $\mathcal{Q}'(t) = |P_j|'(t)$, and $\big||P_j|'\big| \le |P_j'|$ wherever $|P_j|$ is differentiable. This gives the bound for $\mathcal{Q}'$.

*Window.* Fix $\psi \in C_c^\infty(-\frac12,\frac12)$, real, even, nonnegative and not identically 0, let $\check\psi(x) = \int\psi(\omega)e^{i\omega x}d\omega$ and $W = |\check\psi|^2/\check\psi(0)^2$. Then $W \ge 0$ is a Schwartz function with $W(0) = 1$, and $\mathcal{F}W$ is supported in $[-1,1]$. Choose $0 < y_0 \le \frac14$ with $W \ge \frac12$ on $[-y_0, y_0]$ and put $\mathfrak{w}(t) = W(y_0(t - \frac32 T)/(\frac{T}{2}+3))$. Then $\mathfrak{w} \ge \frac12$ on $[T-3, 2T+3]$, $\|\mathfrak{w}\|_4^4 \asymp T$, and $\mathcal{F}\mathfrak{w}$ and $\mathcal{F}(\mathfrak{w}^4)$ are supported in $|\omega| \le 2y_0/T$ and $|\omega| \le 8y_0/T$.

*Proof of* (5.15). Let $f_j = \mathfrak{w}P_j$. Then $\mathcal{F}f_j(\omega) = \sum_{i\le j} p_i^{-1/2}\mathcal{F}\mathfrak{w}(\omega + \log p_i)$, a sum of packets supported in $|\omega + \log p_i| \le 2y_0/T$. Since $\log p_{i+1} - \log p_i \ge \log(1 + 1/X_+) \ge (3/T)^{1/2}/2 > 4y_0/T$ for $T \ge T_0$, the packets are disjoint, and $f_j$ is obtained from $f_J$ by removing the packets of $p_{j+1}, \dots, p_J$, which lie below a point $\omega_j$ separating them from the others: $f_j(t) = f_J(t) - \frac{1}{2\pi}\int_{-\infty}^{\omega_j}\mathcal{F}f_J(\omega)e^{i\omega t}d\omega$. Hence $\max_j|f_j| \le |f_J| + \mathcal{C}f_J$, and on $[T-3, 2T+3]$, $\mathcal{Q} \le 2\max_j|f_j|$. By Theorem 3.11,

$$\int_{T-3}^{2T+3}\mathcal{Q}^4 \le 16\int_{\mathbb{R}}\big(|f_J| + \mathcal{C}f_J\big)^4 \ll \int_{\mathbb{R}}\mathfrak{w}^4|P_J|^4\,dt.$$

Write $P_J^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$ and $\eta_{p_ip_{i'}} = 2(p_ip_{i'})^{-1/2}$ for $i < i'$, all $n \le X_+^2 \le T/3$ (unique factorization makes the representation unique). Expanding, $\int\mathfrak{w}^4|P_J^2|^2 = \sum_{n,n'}\eta_n\eta_{n'}\,\mathcal{F}(\mathfrak{w}^4)(\log(n/n'))$, and $|\log(n/n')| \ge 1/\max(n,n') \ge 3/T > 8y_0/T$ for $n \ne n'$, so only the diagonal remains: $\int\mathfrak{w}^4|P_J|^4 = \|\mathfrak{w}\|_4^4\sum_n\eta_n^2 \ll T\big(\sum_p p^{-2} + 4V_\pm^2\big) \ll TV_\pm^2$, because $\sum_{X_-<p\le X_+}p^{-2} \le V_\pm/X_- \le 4V_\pm/X_+ \le 4V_\pm^2$. The same argument with the polynomials $\sum_{i\le j}\log p_i\, p_i^{-1/2-it}$, whose maximum dominates $|\mathcal{Q}'|$, gives the second bound, with $\eta_n$ multiplied by $\log p_i\log p_{i'} \le (\log X_+)^2$.

*At the ordinates.* Let $T, 2T$ be non-ordinates (otherwise move them by less than the distance to the nearest ordinate, which changes neither side). As Lebesgue–Stieltjes measures

on $(T,2T]$, $dN=\mathcal{L}\,dt+dG$, and $\mathcal{Q}^2$ is Lipschitz, so integration by parts [7, §3.5] gives

$$\sum_{k\in\mathcal{I}_T}\mathcal{Q}(\gamma_k)^2=\int_T^{2T}\mathcal{Q}^2\mathcal{L}\,dt+\Big[\mathcal{Q}^2G\Big]_T^{2T}-2\int_T^{2T}G\,\mathcal{Q}\mathcal{Q}'\,dt, \tag{5.17}$$

multiplicities included. By Cauchy–Schwarz and (5.15) the first term is $\ll \log T\cdot T^{1/2}(TV_\pm^2)^{1/2}=TV_\pm\log T$. The boundary term is $\ll X_+\log T$, since $\mathcal{Q}\le\sum_{p\le X_+}p^{-1/2}\ll X_+^{1/2}$ and $G\ll\log T$. By Hölder's inequality with exponents $2,4,4$, Theorem 3.3 ($\int_T^{2T}G^2\ll T\log\log T$) and (5.15), the last term is $\ll (T\log\log T)^{1/2}(TV_\pm^2)^{1/4}(T(\log X_+)^4V_\pm^2)^{1/4}=T\sqrt{\log\log T}\,(\log X_+)V_\pm$. Divide by $\#\mathcal{I}_T\gg T\log T$.

*At the clock points.* For consecutive indices, $\tau_{k+1}-\tau_k=\int_k^{k+1}dx/\mathcal{L}(\tau_x)\ge 2\pi/\log(3T)=\mathfrak{b}_T$ while $\tau_{k+1}\le 3T$, so the intervals $\mathcal{U}_k=[\tau_k,\tau_k+\mathfrak{b}_T]$, $k\in\mathcal{I}_T^\tau$, are disjoint and contained in $[T,2T+1]$. For an absolutely continuous $g$ on an interval $\mathcal{U}$ of length $\mathfrak{b}$ and $x\in\mathcal{U}$, $|g(x)|\le\frac{1}{\mathfrak{b}}\int_{\mathcal{U}}|g|+\int_{\mathcal{U}}|g'|$, hence $g(x)^2\le\frac{2}{\mathfrak{b}}\int_{\mathcal{U}}g^2+2\mathfrak{b}\int_{\mathcal{U}}g'^2$. With $g=\mathcal{Q}$ and summing, $\sum_{k\in\mathcal{I}_T^\tau}\mathcal{Q}(\tau_k)^2\le\frac{2}{\mathfrak{b}_T}\int_T^{2T+1}\mathcal{Q}^2+2\mathfrak{b}_T\int_T^{2T+1}\mathcal{Q}'^2\ll TV_\pm\log T+\frac{T(\log X_+)^2V_\pm}{\log T}$, by Cauchy–Schwarz and (5.15). Divide by $\#\mathcal{I}_T^\tau\gg T\log T$ and use $\log X_+\le\log T$. □

**Theorem 5.12.** *For $\frac{\log 2}{\log T}\le c\le\frac12$ let $X_k=\max\{2,(\tau_k/2\pi)^c\}$ and $\tilde u_k=s_k-S_{X_k}(\gamma_k)$. Uniformly in $c$,*

$$\Big|\langle\tilde u_k^2\rangle_{\mathcal{I}_T}-\frac{1}{2\pi^2}\log\frac1c\Big|\ll 1+\sqrt{\log(1/c)}. \tag{5.18}$$

*For fixed $c$, $\langle(\tilde u_k-u_k(X_-))^2\rangle_{\mathcal{I}_T}\to 0$, with $X_-$ as in the proof.*

*Proof.* For $T\ge T_0$ and $k\in\mathcal{I}_T$, $|\delta_k|\le 1$ by (3.4), so $\tau_k\in[T-1,2T+1]$ and $X_-\le X_k\le X_+$ with $X_\pm$ the values of $\max\{2,(y/2\pi)^c\}$ at $y=T-1$ and $y=2T+1$. Then $X_+\le((2T+1)/2\pi)^{1/2}\le(T/3)^{1/2}$ and $X_+/X_-\le((2T+1)/(T-1))^{1/2}\le 2$. The prime powers $p^m$ with $X_-<p\le X_k$ are those in $S_{X_k}-S_{X_-}$, so by (2.10)

$$|S_{X_k}(\gamma_k)-S_{X_-}(\gamma_k)|\le\frac1\pi\mathcal{Q}(\gamma_k)+\frac1\pi\sum_{X_-<p\le X_+}\sum_{m\ge2}\frac{1}{mp^{m/2}}\le\frac1\pi\mathcal{Q}(\gamma_k)+1.1\,V_\pm,$$

since the terms $m=1$ are $\frac1\pi\Im\sum_{X_-<p\le X_k}p^{-1/2-i\gamma_k}$ and $\sum_{m\ge2}p^{-m/2}/m\le p^{-1}(1-2^{-1/2})^{-1}$. By Theorem 3.8, $V_\pm=\log\frac{\log X_+}{\log X_-}+O(\frac{1}{\log X_-})\ll\frac{1}{\log X_-}\ll\frac{1}{1+c\log T}$ (and $V_\pm=0$ if $X_-=X_+=2$). Lemma 5.11 gives

$$\langle(\tilde u_k-u_k(X_-))^2\rangle_{\mathcal{I}_T}\ll V_\pm+\frac{(1+c\log T)\sqrt{\log\log T}}{(1+c\log T)\log T}+T^{-1/2}+V_\pm^2\ll 1,$$

which tends to 0 for fixed $c$. Theorem 5.1 with $X=X_-$, for which $\log(1/c_-)=\log(1/c)+O(1)$ with $c_-=\log X_-/\log T$, gives (5.1) for $u_k(X_-)$; (5.18) follows from $|\langle\tilde u^2\rangle-\langle u^2\rangle|\le 2\langle u^2\rangle^{1/2}\langle(\tilde u-u)^2\rangle^{1/2}+\langle(\tilde u-u)^2\rangle$. □

## 6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set $I$ of indices and a prime power $n$, define the response statistics at the zeros and at the clock points,

$$\mathcal{R}_n^\gamma(I)=\frac{2}{a_n}\langle s_k\sin(\gamma_k\log n)\rangle_I,\qquad \mathcal{R}_n^\tau(I)=\frac{2}{a_n}\langle s_k\sin(\tau_k\log n)\rangle_I, \tag{6.1}$$

and, for a set $A$ of prime powers with $\sum_{n\in A}|a_n|<\infty$, the cross term

$$\mathrm{Cr}_A(I)=\langle S_A(\tau_k)^2\rangle_I-v_A. \tag{6.2}$$

For a prime $p$, $2/a_p=-2\pi\sqrt{p}$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; $\mathrm{Cr}_A$ contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

**Theorem 6.1** (Finite-window accounting)**.** *Let $I$ be a finite nonempty set of indices and $A$ a set of prime powers with $\sum_{n\in A}|a_n|<\infty$.*

*(a) For every $k$ and $X\ge 2$,*

$$\begin{aligned}
&r_k(X)=u_k(X)+\Big(S_X(\gamma_k)-S_X(\tau_k)\Big)+\chi_k,\\
&\chi_k=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),\qquad \frac{\delta_k^2}{4\pi\max(\gamma_k,\tau_k)}\le\chi_k\le\frac{\delta_k^2}{4\pi\min(\gamma_k,\tau_k)},\\
&|S_X(\gamma_k)-S_X(\tau_k)|\le M_X|\delta_k|.
\end{aligned} \tag{6.3}$$

*(b) For every real $\lambda$,*

$$\Big\langle s_k^2-(s_k-\lambda S_A(\tau_k))^2\Big\rangle_I=\sum_{n\in A}a_n^2\Big(\lambda\,\mathcal{R}_n^{\tau}(I)-\frac{\lambda^2}{2}\Big)-\lambda^2\,\mathrm{Cr}_A(I). \tag{6.4}$$

*In particular, for the full local factors,*

$$\begin{aligned}
&\langle s_k^2-(s_k-S_X(\tau_k))^2\rangle_I=D_X(I)-\mathrm{Cr}_X(I),\\
&D_X(I)=\sum_{n\in A_X}a_n^2\Big(\mathcal{R}_n^{\tau}(I)-\frac12\Big)=2\langle s_kS_X(\tau_k)\rangle_I-v_X,
\end{aligned} \tag{6.5}$$

*with $\mathrm{Cr}_X=\mathrm{Cr}_{A_X}$.*

*(c) In units of zeros, the gain of the one-step clock over the bare clock is*

$$\begin{aligned}
&\Big\langle(\mathcal{L}(\tau_k)\delta_k)^2-r_k(X)^2\Big\rangle_I=D_X(I)-\mathrm{Cr}_X(I)+2\langle\chi_kS_X(\tau_k)\rangle_I,\\
&|\langle\chi_kS_X(\tau_k)\rangle_I|\le\sigma_X\max_{k\in I}\chi_k.
\end{aligned} \tag{6.6}$$

*Proof.* (a) By (2.12), $r_k(X)=\mathcal{L}(\tau_k)\delta_k-S_X(\tau_k)$. Put $\chi_k=\mathcal{L}(\tau_k)\delta_k-s_k$; since $s_k=N_0(\tau_k)-N_0(\gamma_k)=\int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$\chi_k=\int_{\gamma_k}^{\tau_k}\Big(\mathcal{L}(\tau_k)-\mathcal{L}(y)\Big)dy=\frac{1}{2\pi}\int_{\gamma_k}^{\tau_k}\log\frac{\tau_k}{y}\,dy=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),$$

the last step by $\int\log y\,dy=y\log y-y$. If $\gamma_k<\tau_k$, then $\frac{\tau_k-y}{\tau_k}\le\log\frac{\tau_k}{y}\le\frac{\tau_k-y}{y}\le\frac{\tau_k-y}{\gamma_k}$ for $y\in[\gamma_k,\tau_k]$, and integration gives $\frac{\delta_k^2}{4\pi\tau_k}\le\chi_k\le\frac{\delta_k^2}{4\pi\gamma_k}$; if $\tau_k<\gamma_k$, write $\chi_k=\frac{1}{2\pi}\int_{\tau_k}^{\gamma_k}\log\frac{y}{\tau_k}\,dy$ and use $\frac{y-\tau_k}{\gamma_k}\le\frac{y-\tau_k}{y}\le\log\frac{y}{\tau_k}\le\frac{y-\tau_k}{\tau_k}$; if $\tau_k=\gamma_k$, $\chi_k=0$. Insert $s_k=u_k(X)+S_X(\gamma_k)$. The last bound is the mean value theorem with (2.11).

(b) The series $S_A(\tau_k)=\sum_n a_n\sin(\tau_k\log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since $I$ is finite, the mean over $I$ may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_kS_A(\tau_k)\rangle_I-\lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_kS_A(\tau_k)\rangle_I=\sum_n a_n\langle s_k\sin(\tau_k\log n)\rangle_I=\frac12\sum_n a_n^2\mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I=v_A+\mathrm{Cr}_A(I)$. Taking $\lambda=1$ and $A=A_X$ gives (6.5).

(c) By (a), $\mathcal{L}(\tau_k)\delta_k=s_k+\chi_k$ and $r_k(X)=s_k+\chi_k-S_X(\tau_k)$, so $(\mathcal{L}(\tau_k)\delta_k)^2-r_k(X)^2=2s_kS_X(\tau_k)-S_X(\tau_k)^2+2\chi_kS_X(\tau_k)$; average and use (6.5) and $|S_X|\le\sigma_X$. □

*Remark* 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k\log p$ is assumed. A threshold $\mathcal{R}_p^\tau=\frac12$ describes a single diagonal prime contribution and does not determine the optimal cutoff when $\mathrm{Cr}_X$ and the prime powers matter.

*Remark* 6.3 (Optimal trigonometric correction). For a finite set $A$ of frequencies let $\varsigma_n(k)=\sin(\tau_k\log n)$, $\mathbf{Q}=(\langle\varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b}=(\langle s_k\varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol\lambda}=\sum_n\lambda_n\varsigma_n$. Ordinary least squares gives $\langle(s_k-\Upsilon_{\boldsymbol\lambda}(k))^2\rangle_I=\langle s_k^2\rangle_I-2\boldsymbol\lambda^{\mathsf T}\mathbf{b}+\boldsymbol\lambda^{\mathsf T}\mathbf{Q}\boldsymbol\lambda$, minimized by $\boldsymbol\lambda=\mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf T}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n=\frac12 a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n=a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau\ne1$ and $\mathrm{Cr}_A\ne0$.

*Remark* 6.4 (Clock point against zero). For a prime power $n$, $\sin(\gamma_k\ell_n)-\sin(\tau_k\ell_n)=\int_0^1\frac{d}{dy}\sin((\tau_k-y\delta_k)\ell_n)\,dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^\gamma(I)-\mathcal{R}_n^\tau(I)=-\frac{2\ell_n}{a_n}\int_0^1\Big\langle s_k\delta_k\cos\Big((\tau_k-y\delta_k)\ell_n\Big)\Big\rangle_I\,dy. \tag{6.7}$$

For fixed $n$ and $I=\mathcal{I}_T$ the right side is $O(\langle|s_k\delta_k|\rangle_{\mathcal{I}_T})=O(\log\log T/\log T)$ by (2.6) and Remark 3.15; for $\ell_n=\theta L_T$ with fixed $\theta$ the phase change $\delta_k\ell_n$ has mean square $\theta^2L_T^2\langle\delta_k^2\rangle_{\mathcal{I}_T}\sim2\theta^2\log\log T$, so it is not small, and the transport integral need not be small; no asymptotic statement about this difference is proved.

*Remark* 6.5 (Fixed prime sets). For fixed $X$, $|\delta_k^X-\delta_k|\le\sigma_X/\mathcal{L}(\tau_k)\to0$ and $|u_k(X)-s_k|\le\sigma_X$. A fixed set of primes changes neither $\limsup_k|\delta_k|$ nor the unboundedness of the defect.

## 7. Theorem D: a computable clock that approaches the zeros

The one-step clock $\tau_k^X=\tau_k-y_k$, $y_k=S_X(\tau_k)/\mathcal{L}(\tau_k)$, uses only $\tau_k$ and the primes $p\le X$. By Theorem 6.1(a),

$$r_k(X)=u_k(X)+\mathfrak{t}_k(X)+\chi_k,\qquad \mathfrak{t}_k(X)=S_X(\gamma_k)-S_X(\tau_k), \tag{7.1}$$

where Theorem 5.1 controls $u_k(X)$ and $0\le\chi_k\ll1/T$ on $\mathcal{I}_T$. This section controls the transport term $\mathfrak{t}_k(X)$ in mean square, with no assumption on the joint behaviour of $\delta_k$ and the phases $\tau_k\log p$.

**Lemma 7.1.** *For $T\ge T_0$ and $2\le X\le T^{1/2}$,*

$$\int_{T-3}^{2T+3}S_X'(t)^4\,dt\ll T(1+\log X)^4,\qquad \int_{T-3}^{2T+3}S_X''(t)^4\,dt\ll T(1+\log X)^8.$$

*Proof.* By Section 2.4, the series

$$S'_X(t)=\sum_{n\in A_X} a_n\log n\cos(t\log n),\qquad S''_X(t)=-\sum_{n\in A_X} a_n\log^2 n\sin(t\log n)$$

converge absolutely and uniformly. Split according to $m=1$, $m=2$, $m\ge 3$ in $n=p^m$ and use $(y_1+y_2+y_3)^4\le 27(y_1^4+y_2^4+y_3^4)$. For $\nu\in\{1,2\}$ the terms $m\ge 3$ of the $\nu$-th derivative are bounded by $\frac{1}{\pi}\sum_p\sum_{m\ge3}\frac{(m\log p)^\nu}{mp^{m/2}}<\infty$, uniformly in $X$ and $t$. The terms $m=1$ are bounded by $\frac{1}{\pi}|\mathcal{D}_\nu(t)|$ with $\mathcal{D}_\nu(t)=\sum_{p\le X}\log^\nu p\;p^{-1/2-it}$; $\mathcal{D}_\nu^2=\sum_n\eta_n n^{-it}$ over $n=pp'\le X^2\le T$ with $\sum_n\eta_n^2\le 2(\sum_{p\le X}\log^{2\nu}p/p)^2\ll(1+\log X)^{4\nu}$ (Theorem 3.8 and partial summation), so Theorem 3.7 on an interval of length $T+6$ gives $\int|\mathcal{D}_\nu|^4=\sum\eta_n^2(T+6+O(n))\ll T(1+\log X)^{4\nu}$. The terms $m=2$ are bounded by $\frac{1}{2\pi}|\mathcal{D}'_\nu(t)|$ with $\mathcal{D}'_\nu(t)=\sum_{p\le X}(2\log p)^\nu p^{-1-2it}$; here $\mathcal{D}'^2_\nu=\sum_n\eta'_n n^{-it}$ over $n=(pp')^2\le X^4$, with $\sum_n\eta'^2_n\ll 1$ and $\sum_n n\eta'^2_n\ll(\sum_{p\le X}\log^{2\nu}p)^2\ll X^2(1+\log X)^{4\nu}\le T(1+\log X)^{4\nu}$, so again $\int|\mathcal{D}'_\nu|^4\ll T(1+\log X)^{4\nu}$. □

**Lemma 7.2** (Transport)**.** *For $T\ge T_0$ and $2\le X\le T^{1/2}$,*

$$\big\langle \mathfrak{t}_k(X)^2\big\rangle_{\mathcal{I}_T}\ll\frac{\log\log T\,(1+\log X)^2}{\log^2 T}. \tag{7.2}$$

*Proof. Fourth moment of $\delta_k$.* For $T\ge T_0$ and $k\in\mathcal{I}_T$, $|\delta_k|\le 1$ by (3.4); so the point $\xi_k$ of (2.6) is $\ge T-1$ and $\mathcal{L}(\xi_k)\gg\log T$. By (3.3) with $j=2$,

$$\langle\delta_k^4\rangle_{\mathcal{I}_T}=\langle s_k^4\mathcal{L}(\xi_k)^{-4}\rangle_{\mathcal{I}_T}\ll\frac{(\log\log T)^2}{\log^4 T}. \tag{7.3}$$

*Maximal function.* Let $\varphi\in C^\infty(\mathbb{R})$ with $0\le\varphi\le 1$, $\varphi=1$ on $[T-2,2T+2]$, $\varphi=0$ outside $[T-3,2T+3]$ and $|\varphi'|\le 2$, and let $\mathfrak{h}=\varphi S'_X$. For $k\in\mathcal{I}_T$ the segment between $\gamma_k$ and $\tau_k$ lies in $[T-1,2T+1]$, where $\mathfrak{h}=S'_X$, so

$$|\mathfrak{t}_k(X)|=\Big|\int_{\tau_k}^{\gamma_k}\mathfrak{h}(y)\,dy\Big|\le\int_{\tau_k-|\delta_k|}^{\tau_k+|\delta_k|}|\mathfrak{h}|\le 2|\delta_k|\,\mathcal{M}\mathfrak{h}(\tau_k) \tag{7.4}$$

(trivially if $\delta_k=0$). The function $\mathfrak{h}$ is $C^\infty$ with compact support. By Theorem 3.10, $g=\mathcal{M}\mathfrak{h}$ is locally absolutely continuous with $\|g\|_4\ll\|\mathfrak{h}\|_4$ and $\|g'\|_4\le\|\mathcal{M}(\mathfrak{h}')\|_4\ll\|\mathfrak{h}'\|_4$; by Lemma 7.1, $\|\mathfrak{h}\|_4^4\ll T(1+\log X)^4$ and, since $\mathfrak{h}'=\varphi' S'_X+\varphi S''_X$, $\|\mathfrak{h}'\|_4^4\ll T(1+\log X)^8$.

*Sampling at the clock points.* As in the proof of Lemma 5.11, the intervals $\mathcal{U}_k=[\tau_k,\tau_k+\mathfrak{b}_T]$, $k\in\mathcal{I}_T$, are disjoint and contained in $[T-1,2T+2]$. For $x\in\mathcal{U}$, an interval of length $\mathfrak{b}$, $|g(x)|\le\frac{1}{\mathfrak{b}}\int_{\mathcal{U}}|g|+\int_{\mathcal{U}}|g'|$; by Hölder's inequality and $(y_1+y_2)^4\le 8(y_1^4+y_2^4)$, $g(x)^4\le\frac{8}{\mathfrak{b}}\int_{\mathcal{U}}g^4+8\mathfrak{b}^3\int_{\mathcal{U}}|g'|^4$. Summing over $k\in\mathcal{I}_T$ and dividing by $\#\mathcal{I}_T\gg T\log T$,

$$\langle g(\tau_k)^4\rangle_{\mathcal{I}_T}\ll\frac{1}{T}\int_{\mathbb{R}}g^4+\frac{1}{T\log^4 T}\int_{\mathbb{R}}|g'|^4\ll(1+\log X)^4+\frac{(1+\log X)^8}{\log^4 T}\ll(1+\log X)^4.$$

*Conclusion.* By (7.4) and Cauchy–Schwarz, $\langle\mathfrak{t}_k^2\rangle\le 4\langle\delta_k^2 g(\tau_k)^2\rangle\le 4\langle\delta_k^4\rangle^{1/2}\langle g(\tau_k)^4\rangle^{1/2}$; insert (7.3). □

**Theorem 7.3** (Theorem D)**.** *There are absolute constants $T_0$ and $C$ such that for $T \ge T_0$ and $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,*

$$\Big|\big\langle r_k(X)^2\big\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \le C\Big(1+\sqrt{\log(1/c)}\Big)\Big(1+c^2\log\log T\Big). \tag{7.5}$$

*Proof.* By Theorem 6.1(a), $0 \le \chi_k \le \delta_k^2/(4\pi(T-1)) \le 1/T$ on $\mathcal{I}_T$. Since $1+\log X \le (1+\frac{1}{\log 2})c\log T$, Lemma 7.2 gives $\langle \mathfrak{t}_k^2\rangle \ll c^2\log\log T$, so $d_k = r_k(X) - u_k(X) = \mathfrak{t}_k + \chi_k$ has $\langle d_k^2\rangle \ll c^2\log\log T + T^{-2}$. Theorem 5.1 gives $\langle u_k(X)^2\rangle \ll 1+\log(1/c)$. Then $|\langle r_k^2\rangle - \langle u_k^2\rangle| \le 2\langle u_k^2\rangle^{1/2}\langle d_k^2\rangle^{1/2} + \langle d_k^2\rangle \ll (1+\sqrt{\log(1/c)})(c\sqrt{\log\log T}+T^{-1}) + c^2\log\log T$, which is bounded by the right side of (7.5) because $y \le 1+y^2$. Combine with (5.1). □

**Corollary 7.4.** *Let $X = T^{1/\sqrt{\log\log T}}$. Then*

$$\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}\log\log\log T + O\Big(\sqrt{\log\log\log T}\Big), \tag{7.6}$$

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(\tau_k^X - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}, \tag{7.7}$$

*while $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}(\tau_k-\gamma_k)^2 = (2+o(1))\log\log T/\log^2 T$; the ratio is asymptotic to $\log\log\log T/(2\log\log T)$.*

*Proof.* Here $c = (\log\log T)^{-1/2}$, $2 \le X \le T^{1/2}$ for $T \ge T_0$, $\log(1/c) = \frac12\log\log\log T$ and $c^2\log\log T = 1$, so (7.5) gives (7.6). On $\mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$; divide $r_k^2$ by $\mathcal{L}(\tau_k)^2$. For the bare clock, $\delta_k = s_k/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$, and $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ by (3.3). □

**Corollary 7.5** (A cutoff read off the clock)**.** *For $\tau_k > e^e$ let $X_k^{\mathrm{loc}} = \tau_k^{1/\sqrt{\log\log\tau_k}}$ and $\tau_k^{\mathrm{loc}} = \tau_k - S_{X_k^{\mathrm{loc}}}(\tau_k)/\mathcal{L}(\tau_k)$. Then*

$$\frac{1}{\#\mathcal{I}_T^\tau}\sum_{k\in\mathcal{I}_T^\tau}\Big(\tau_k^{\mathrm{loc}} - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}.$$

*Proof.* Let $X_0 = T^{1/\sqrt{\log\log T}}$ and $X_0' = (2T)^{1/\sqrt{\log\log 2T}}$. The function $y \mapsto y/\sqrt{\log y}$ is increasing for $y > e^{1/2}$, so $X_0 \le X_k^{\mathrm{loc}} \le X_0'$ for $k \in \mathcal{I}_T^\tau$, and $\log X_0' - \log X_0 \le \log 2/\sqrt{\log\log T} = o(1)$; hence $X_0' \le 2X_0$ and $X_0' \le (T/3)^{1/2}$ for $T \ge T_0$. As in the proof of Theorem 5.12, $|S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k)| \le \frac{1}{\pi}\mathcal{Q}(\tau_k) + 1.1V_\pm$ with $X_\pm$ replaced by $X_0, X_0'$, and Lemma 5.11 gives a mean square over $\mathcal{I}_T^\tau$ that is $\ll V_\pm \ll 1/\log X_0 = \sqrt{\log\log T}/\log T$. The sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau$ differ by $O(\log T)$ indices, those with $\gamma_k$ or $\tau_k$ within distance 1 of $T$ or $2T$, and for them $|r_k(X_0)| \le |s_k| + \chi_k + \sigma_{X_0} \ll \log T + X_0^{1/2}$ (Lemma 2.2); so the means of $r_k(X_0)^2$ over the two sets differ by $O(T^{-1/2})$, and (7.6) holds over $\mathcal{I}_T^\tau$. Finally $\mathcal{L}(\tau_k)(\tau_k^{\mathrm{loc}} - \gamma_k) = r_k(X_0) - (S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k))$; conclude with Cauchy–Schwarz and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$. □

**Proposition 7.6** (The root of the corrected equation in the monotone regime)**.** *Let $X = X(T) \ge 2$ with $M_X \le \frac12\mathcal{L}(T)$, and let $\mathcal{I}_T'$ be the set of $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac12 \le F_X(2T)$,*

*so that the root $t_k^X \in [T, 2T]$ of Remark 5.7 exists. Then $X \ll \log^2 T$, $\#(\mathcal{I}_T \setminus \mathcal{I}'_T) \ll \log T$, and*

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}'_T} \big(t_k^X - \gamma_k\big)^2 \geq \frac{4}{9\mathcal{L}(2T)^2} \Big( \frac{1}{2\pi^2} \log\log T - O\Big(\sqrt{\log\log T}\Big) \Big) \gg \frac{\log\log T}{\log^2 T}.$$

*Proof.* $M_X \geq \frac{1}{\pi} \sum_{p \leq X} \log p/\sqrt{p} \gg \sqrt{X}$ by Chebyshev's estimate, so $X \ll \mathcal{L}(T)^2 \ll \log^2 T$ and $c = \log X/\log T \leq (2\log\log T + O(1))/\log T$. On $[T, 2T]$, $0 < F'_X \leq \mathcal{L} + M_X \leq \frac{3}{2}\mathcal{L}(2T)$, so $|u_k(X)| = |F_X(t_k^X) - F_X(\gamma_k)| \leq \frac{3}{2}\mathcal{L}(2T)|t_k^X - \gamma_k|$ for $k \in \mathcal{I}'_T$. An index $k \in \mathcal{I}_T$ is missing from $\mathcal{I}'_T$ only if $k - \frac{1}{2} < F_X(T)$ or $k - \frac{1}{2} > F_X(2T)$; since $N(T) < k \leq N(2T)$ and $|F_X - N| \leq |G| + \sigma_X \ll \log T$ at $T$ and $2T$, there are $O(\log T)$ such $k$, each with $|u_k(X)| \ll \log T$ (Lemma 2.2). Hence $\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}'_T} u_k(X)^2 = \langle u_k(X)^2 \rangle_{\mathcal{I}_T} + O(\frac{\log^3 T}{T \log T})$, and Theorem 5.1 with $\log(1/c) \geq \log\log T - \log(2\log\log T + O(1))$ bounds it from below. □

Thus a root of the corrected equation can be selected only where it gains nothing in order: the error in height of $t_k^X$ has mean square of the same order $\log\log T/\log^2 T$ as that of $\tau_k$. Theorem 7.3 avoids roots and takes one Newton step instead.

*Remark* 7.7 (The Newton term and the restriction on $c$). Let $\xi_k^X$ be a point between $\gamma_k$ and $\tau_k$ with $\mathfrak{t}_k(X) = -S'_X(\xi_k^X)\delta_k$ (mean value theorem). Since $\delta_k = (r_k(X) + S_X(\tau_k))/\mathcal{L}(\tau_k)$ by (2.12), (7.1) becomes the exact relation

$$r_k(X)\Big(1 + \frac{S'_X(\xi_k^X)}{\mathcal{L}(\tau_k)}\Big) = u_k(X) + \chi_k - y_k\, S'_X(\xi_k^X). \tag{7.8}$$

The term $y_k S'_X(\xi_k^X)$ is the second-order term of Newton's method. Heuristically, $\langle S_X^2 \rangle \approx v_X \approx \frac{1}{2\pi^2}\log\log X$ and $\langle S_X'^2 \rangle/\mathcal{L}(T)^2 \approx \sum_{p \leq X} \frac{\log^2 p}{2\pi^2 p}/\mathcal{L}(T)^2 \approx c^2$, so this term has mean square of order $c^2 \log\log T$ if the two factors decorrelate: the order of Lemma 7.2 is the order of the Newton term, and the lemma cannot be improved by bounds that ignore the correlation between $u_k(X)$ and $\mathfrak{t}_k(X)$. Whether the transport cancels against $u_k(X)$ is open; a proof of such a cancellation would allow a fixed $c$ in Corollary 7.4 (Question 10.2). The denominator in (7.8) may vanish where $F'_X(\tau_k) \leq 0$.

## 8. Iterating the Newton step

Theorem 7.3 takes one Newton step from $\tau_k$. Its restriction $c \leq (\log\log T)^{-1/2}$ comes from the transport over the interval between $\tau_k$ and $\gamma_k$, whose length is that of $\delta_k$. After one step the error is smaller, so a second step can use a larger cutoff. This section proves the second step and then every further step, with constants uniform in the number of steps (Section 8.5).

*8.1. The exact recursion*

For a sequence of cutoffs $X_1, X_2, \dots \geq 2$ define $x_k^{(0)} = \tau_k$ and

$$x_k^{(j)} = x_k^{(j-1)} - \frac{N_0(x_k^{(j-1)}) + S_{X_j}(x_k^{(j-1)}) - (k - \frac{1}{2})}{\mathcal{L}(x_k^{(j-1)})}, \qquad e_{j,k} = x_k^{(j)} - \gamma_k, \tag{8.1}$$

so that $x_k^{(1)} = \tau_k^{X_1}$ and $e_{0,k} = \delta_k$. For a finite index set $I$ and $r \geq 1$ let $\|a\|_{r,I} = \langle |a_k|^r \rangle_I^{1/r}$.

**Proposition 8.1** (Exact recursion)**.** *Let $j \ge 1$ and suppose $x_k^{(j-1)} > 2\pi$. Then*

$$\mathcal{L}(x_k^{(j-1)})\, e_{j,k} = u_k(X_j) + \Big(S_{X_j}(\gamma_k) - S_{X_j}(x_k^{(j-1)})\Big) + \chi_{j,k}, \qquad 0 \le \chi_{j,k} \le \frac{e_{j-1,k}^2}{4\pi \min(\gamma_k, x_k^{(j-1)})}. \tag{8.2}$$

*Let $\varphi_j \in C^\infty(\mathbb{R})$, $0 \le \varphi_j \le 1$, be $1$ on an interval containing all segments between $\gamma_k$ and $x_k^{(j-1)}$, $k \in I$, and $g_j = \mathcal{M}(\varphi_j S'_{X_j})$. Then for every $r \ge 1$,*

$$\begin{aligned} \Big\|\mathcal{L}(x^{(j-1)})e_j\Big\|_{2r,I} &\le \|u(X_j)\|_{2r,I} + 2\,\|e_{j-1}\|_{4r,I}\,\|g_j(x^{(j-1)})\|_{4r,I} + \|\chi_j\|_{2r,I}, \\ \|\chi_j\|_{2r,I} &\le \frac{\|e_{j-1}\|_{4r,I}^2}{4\pi \min_{k\in I}\min(\gamma_k, x_k^{(j-1)})}. \end{aligned} \tag{8.3}$$

*Proof.* Write $x = x_k^{(j-1)}$. By (8.1), $\mathcal{L}(x)e_{j,k} = \mathcal{L}(x)(x-\gamma_k) - N_0(x) - S_{X_j}(x) + k - \frac12$. As in the proof of Theorem 6.1(a), $N_0(x) - N_0(\gamma_k) = \int_{\gamma_k}^x \mathcal{L} = \mathcal{L}(x)(x-\gamma_k) - \chi_{j,k}$ with $\chi_{j,k} = \int_{\gamma_k}^x (\mathcal{L}(x) - \mathcal{L}(y))\,dy$, which lies between $(x-\gamma_k)^2/(4\pi\max(\gamma_k,x))$ and $(x-\gamma_k)^2/(4\pi\min(\gamma_k,x))$ by the same estimates of log. Hence $\mathcal{L}(x)e_{j,k} = k - \frac12 - N_0(\gamma_k) - S_{X_j}(x) + \chi_{j,k} = u_k(X_j) + S_{X_j}(\gamma_k) - S_{X_j}(x) + \chi_{j,k}$. For (8.3), $|S_{X_j}(\gamma_k) - S_{X_j}(x)| \le \int_{x-|e_{j-1,k}|}^{x+|e_{j-1,k}|} |\varphi_j S'_{X_j}| \le 2|e_{j-1,k}|\, g_j(x)$, and Minkowski's and Hölder's inequalities in $\ell^{2r}(I)$. □

The recursion (8.3) is exact and uses no independence between the errors and the prime phases. It reduces the error of each step to three quantities: the moments of the defect $u_k(X_j)$ at the true ordinates (Theorem 5.1 and its higher-moment versions), the moments of the previous error, and the moments of the maximal function $g_j$ sampled at the computable points $x_k^{(j-1)}$. The last is the only new difficulty: the points $x_k^{(j-1)}$ are not separated like the $\tau_k$.

*8.2. Moments of order $2r$ of the defect*

**Lemma 8.2.** *There is an absolute constant $C$ such that for every integer $r \ge 1$, every $T$ with $T^{1/(160r)} \ge 2$ and every $2 \le X \le T^{1/(160r)}$, with $V$ as in Lemma 5.8 computed with $Y_{2r} = T^{1/(160r)}$ in place of $Y_0$,*

$$\int_T^{2T} E_X^{4r}\,dt \le \Big(C(r^2 + \sqrt{r(1+V)})\Big)^{4r} T, \qquad \langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le \Big(C(r^2 + \sqrt{r(1+V)})\Big)^{2r},$$

*and for every $r \ge 1$ and $2 \le X \le T^{1/(2r)}$,*

$$\int_{T-3}^{2T+3} |S_X^{(\nu)}(t)|^{4r} dt \le (Cr)^{2r}\, T(1+\log X)^{4r\nu} \qquad (\nu = 1, 2).$$

*Proof.* The first bound: $E_X = E_0 + \mathcal{W}$ with $E_0 = S - S_{Y_{2r}}^{\mathrm{pr}}$, $\|E_0\|_{L^{4r}(T,2T)} \le C(2r)^2 T^{1/4r}$ by Theorem 3.5 with $k = 2r$, and $\mathcal{W}$ as in Lemma 5.10: $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$, $\mathcal{D}_1^{2r} = \sum_n \eta_n n^{-it}$ over products of $2r$ primes $\le \max(X, Y_{2r})$, so $n \le T^{1/80}$, with $\sum |\eta_n|^2 \le (2r)!\,(V+1)^{2r}$ (each $n$ has at most $(2r)!$ ordered factorizations, so $\sum_n |\eta_n|^2 \le (2r)! \sum_{p_1,\dots,p_{2r}} \prod p_i^{-1}$); Theorem 3.7 gives $\int |\mathcal{D}_1|^{4r} \le 2T \sum |\eta_n|^2$, whence $\|\mathcal{W}_1\|_{L^{4r}} \le C\sqrt{r}\sqrt{1+V}\, T^{1/4r}$ by $((2r)!)^{1/4r} \le \sqrt{2r}$. The terms $m = 2$ and $m \ge 3$ are $O(T^{1/4r})$ in the same norm (products of $2r$ squares of primes,

$\sum|\eta_n'|^2 \le (2r)!\,(\sum_p p^{-2})^{2r} \le (2r)!$, and a bounded function). Minkowski's inequality in $L^{4r}$ gives the first bound. The second: Theorem 4.1 with $j=r$ and $F=F_X$, $\Psi_r(u)\ge u^{2r}$, gives $\sum_{k\in\mathcal{I}_T} u_k^{2r} \le \int_T^{2T}(|E_X|+T^{-1})^{2r}(\mathcal{L}+|S_X'|)\,dt + O\big(4^rT^{1/100}+(2\log T)^{2r+1}\big)$, the last term bounding the boundary terms and the end pieces, since $|u_k(X)|\le \log T+\sqrt{X}$ (Lemma 2.2 and (2.11)) and $X^{r+1/2}\le T^{1/100}$. With $\mathcal{E}_{4r}=\int_T^{2T}E_X^{4r}$, Hölder's inequality gives $\int_T^{2T}E_X^{2r}\le T^{1/2}\mathcal{E}_{4r}^{1/2}$ and, by (5.8), $\int_T^{2T}E_X^{2r}|S_X'|\le \mathcal{E}_{4r}^{1/2}C_0T^{1/2}(1+\log X)$; so the integral is at most $(\mathcal{L}(2T)+C_0(1+\log X))T^{1/2}(\mathcal{E}_{4r}^{1/2}+T^{-2r+1/2})$. Dividing by $\#\mathcal{I}_T\ge\frac12T\mathcal{L}(T)$ and using $\mathcal{L}(2T)\le 2\mathcal{L}(T)$ and $1+\log X\le 2\pi\mathcal{L}(T)$, $\langle u_k(X)^{2r}\rangle_{\mathcal{I}_T}\le(4+4\pi C_0)T^{-1/2}\mathcal{E}_{4r}^{1/2}+o(1)$, and the first bound gives the second. The third: Lemma 7.1 with products of $2r$ primes, $n\le X^{2r}\le T$, and $\sum\eta_n^2\le(2r)!\,(\sum_{p\le X}\log^{2\nu}p/p)^{2r}\le(Cr)^{2r}(1+\log X)^{4r\nu}$, by Theorem 3.8 and partial summation; the terms $m=2$ and $m\ge3$ as in Lemma 7.1. □

The dependence on $r$ matters in Theorem 8.10, where $r$ grows with the number of Newton steps; for fixed $r$ the three bounds read $\ll_r T(1+V)^{2r}$, $\ll_r(1+V)^r$ and $\ll_r T(1+\log X)^{4r\nu}$.

*8.3. Multiplicity of the one-step clock points*

Let $X_1=T^{c_1}$ with $c_1=(\log\log T)^{-1/2}$, $x_k^{(1)}=\tau_k^{X_1}=\tau_k-y_k$ with $y_k=S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$, $\mathfrak{b}_T=2\pi/\log(3T)$ as in Lemma 5.11, and

$$\tilde n(u)=\#\{k\in\mathcal{I}_T:\ u-\mathfrak{b}_T\le x_k^{(1)}\le u\}\qquad(u\in\mathbb{R}).$$

Note that $|y_k|\le\sigma_{X_1}/\mathcal{L}(\tau_k)\ll T^{c_1/2}$, so $x_k^{(1)}\in[T-T^{1/2},2T+T^{1/2}]$ for $T\ge T_0$.

**Lemma 8.3** (Multiplicity)**.** $\displaystyle\int_{\mathbb{R}}\tilde n(u)^2\,du=\sum_{j,k\in\mathcal{I}_T}\Big(\mathfrak{b}_T-|x_j^{(1)}-x_k^{(1)}|\Big)_+\ll\mathfrak{b}_T\,\#\mathcal{I}_T.$

*Proof.* The identity: $\tilde n=\sum_k\mathbf{1}_{[x_k^{(1)},x_k^{(1)}+\mathfrak{b}_T]}$, and the measure of the intersection of two such intervals is $(\mathfrak{b}_T-|x_j^{(1)}-x_k^{(1)}|)_+$. Write $b=\mathfrak{b}_T$ and $M=\#\mathcal{I}_T$. The pairs with $|j-k|\le1$ contribute at most $3bM$. Let $j=k+m$ with $m\ge2$ and $|\rho_{k+m}-x_k^{(1)}|\le b$. Since $\tau_{k+m}-\tau_k\ge mb$ (proof of Lemma 5.11), this forces

$$|y_{k+m}-y_k|\ge(m-1)b.\tag{8.4}$$

Two consequences of (8.4) are counted separately; a close pair at distance $m$ satisfies both.

*Variation.* $y=S_{X_1}/\mathcal{L}$ is $C^\infty$ with $y'=S_{X_1}'/\mathcal{L}-S_{X_1}\mathcal{L}'/\mathcal{L}^2$, and on $J_{k,m}=[\tau_k,\tau_{k+m}]$, of length $|J_{k,m}|\le mb(1+o(1))$,

$$|y_{k+m}-y_k|\le\frac{1+o(1)}{\mathcal{L}(T)}\int_{J_{k,m}}|S_{X_1}'|+\frac{\sigma_{X_1}|J_{k,m}|}{2\pi(T-1)\mathcal{L}(T)^2},$$

and the last term is $\ll T^{c_1/2-1}mb/\log^2T=o((m-1)b)$ uniformly in $m\ge2$. As $b\mathcal{L}(T)=1-o(1)$, (8.4) gives $\frac{1}{|J_{k,m}|}\int_{J_{k,m}}|S_{X_1}'|\ge\frac13\mathcal{L}(T)$ for $T\ge T_0$. Each $t$ lies in at most $m$ of the intervals $J_{k,m}$ ($k$ varying), and $|J_{k,m}|\ge mb$; by Chebyshev's inequality and Jensen's inequality,

$$\#\{k:\text{variation}\}\le\frac{9}{\mathcal{L}(T)^2}\sum_k\frac{1}{|J_{k,m}|}\int_{J_{k,m}}|S_{X_1}'|^2\le\frac{9}{\mathcal{L}(T)^2\,b}\int_{T-1}^{2T+1}|S_{X_1}'|^2\ll T\log T\,c_1^2\asymp M\,c_1^2,$$

by (5.8), uniformly in $m$.

*Size.* (8.4) gives $\max(|y_k|,|y_{k+m}|)\ge\frac{m-1}{2}b$, hence $\max(|S_{X_1}(\tau_k)|,|S_{X_1}(\tau_{k+m})|)\ge\frac{m-1}{3}$ for $T\ge T_0$. Sampling at the clock points as in the proof of Lemma 5.11 (Sobolev on the disjoint intervals $[\tau_k,\tau_k+b]$), $\langle S_{X_1}(\tau_k)^4\rangle_{\mathcal{I}_T}\ll\frac{1}{T}\int S_{X_1}^4+\frac{1}{T\log^4 T}\int|S'_{X_1}|^4\ll(1+\log\log T)^2$, the first integral by Theorem 3.7 with products of two primes ($\sum_{p\le X_1}p^{-1}\ll\log\log T$) and the second by Lemma 7.1. By Chebyshev's inequality of order 4, $\#\{k:\text{size}\}\ll M(\log\log T)^2/(m-1)^4$.

*Sum.* With $H_0=\log\log T$ and $c_1^2=H_0^{-1}$, the number of close ordered pairs at distance $m\ge2$ is $\ll M\min(H_0^{-1},H_0^2(m-1)^{-4})$, and

$$\sum_{m\ge2}\min\Big(H_0^{-1},\frac{H_0^2}{(m-1)^4}\Big)\le\sum_{m-1\le H_0^{3/4}}H_0^{-1}+\sum_{m-1>H_0^{3/4}}\frac{H_0^2}{(m-1)^4}\ll H_0^{-1/4}.$$

Hence the number of pairs $(j,k)$ with $|x_j^{(1)}-x_k^{(1)}|\le b$ is $\ll M$, and each contributes at most $b$. □

**Lemma 8.4** (Sampling at points of bounded multiplicity)**.** *Let $\mathsf{x}_k$, $k\in\mathcal{I}_T$, be real points of $[T-2T^{1/2},2T+2T^{1/2}]$ with multiplicity function $\tilde n(u)=\#\{k\in\mathcal{I}_T:u-\mathfrak{b}_T\le\mathsf{x}_k\le u\}$ satisfying $\int_{\mathbb{R}}\tilde n^2\le\kappa\,\mathfrak{b}_T\#\mathcal{I}_T$ for some $\kappa\ge1$. Let $r\ge1$ be an integer, $2\le X\le T^{1/(4r)}$, $\varphi$ a cutoff equal to 1 on $[T-2T^{1/2},2T+2T^{1/2}]$ as in Lemma 7.2, and $g=\mathcal{M}(\varphi S'_X)$. Then, with an absolute constant $C$,*

$$\|g(\mathsf{x}_k)\|_{4r,\mathcal{I}_T}\le C\,\kappa^{1/(8r)}\sqrt{r}\,(1+\log X).$$

*In particular $\langle g(x_k^{(1)})^4\rangle_{\mathcal{I}_T}\ll(1+\log X)^4$ at the one-step clock points $x_k^{(1)}$, for $2\le X\le T^{1/4}$.*

*Proof.* Let $F=g^{4r}$, locally absolutely continuous with $F'=4rg^{4r-1}g'$ (Theorem 3.10). For $x\in[u-b,u]$, $F(x)\le\frac1b\int_{u-b}^u F+\int_{u-b}^u|F'|$. Averaging over $u\in[\mathsf{x}_k,\mathsf{x}_k+b]$ and summing over $k$,

$$\sum_{k\in\mathcal{I}_T}F(\mathsf{x}_k)\le\frac1b\int_{\mathbb{R}}\tilde n(u)\Big(\frac1b\int_{u-b}^u F+\int_{u-b}^u|F'|\Big)du\le\frac1b\|\tilde n\|_2\Big(\|F\|_2+b\|F'\|_2\Big),$$

by Cauchy–Schwarz and because the averaging operator $h\mapsto\frac1b\int_{u-b}^u h$ has norm $\le1$ on $L^2(\mathbb{R})$. Here $\|F\|_2=\|g\|_{8r}^{4r}\le(A_{8r}\|\varphi S'_X\|_{8r})^{4r}\le(C\sqrt{r}(1+\log X))^{4r}T^{1/2}$ and $\|F'\|_2\le4r\|g\|_{8r}^{4r-1}\|g'\|_{8r}\le4r(1+\log X)\,(C\sqrt{r}(1+\log X))^{4r}T^{1/2}$, by Theorem 3.10 ($p=8r$, $A_p$ bounded) and Lemma 8.2 (third bound with $2r$ in place of $r$, which needs $X^{4r}\le T$). Since $b\cdot4r(1+\log X)\le8\pi r\,c\,(1+o(1))\le7$ for $c=\log X/\log T\le1/(4r)$, we get $\sum_k F(\mathsf{x}_k)\le8\,b^{-1}(\kappa bM)^{1/2}T^{1/2}(C\sqrt{r}(1+\log X))^{4r}\le C'\sqrt{\kappa}\,M\,(C\sqrt{r}(1+\log X))^{4r}$, since $b^{-1}T\asymp M$; take the $4r$-th root. For the $x_k^{(1)}$, $\kappa\ll1$ by Lemma 8.3. □

*8.4. The second step*

**Theorem 8.5** (Second Newton step)**.** *Let $H_0=\log\log T$, $H_1=\log H_0$, $H_2=\log H_1$, $X_1=T^{H_0^{-1/2}}$, $X_2=T^{H_1^{-1/2}}$, and let $x_k^{(2)}$ be given by* (8.1)*. Then*

$$\begin{aligned}&\Big\langle\mathcal{L}(x_k^{(1)})^2e_{2,k}^2\Big\rangle_{\mathcal{I}_T}=\frac{1}{4\pi^2}H_2+O\Big(1+\sqrt{H_2}\Big),\\&\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(2)}-\gamma_k\Big)^2=\frac{\log\log\log\log T+O\Big(\sqrt{\log\log\log\log T}\Big)}{\log^2T}.\end{aligned}\tag{8.5}$$

*Proof. Fourth moment of $e_1$.* By (8.2) with $j=1$ (that is (7.1)), $\mathcal{L}(\tau_k)e_{1,k} = u_k(X_1)+\mathfrak{t}_k(X_1)+\chi_k$. Lemma 8.2 with $r=2$ gives $\langle u_k(X_1)^4\rangle \ll (1+\log(1/c_1))^2 \ll H_1^2$. For the transport, $|\mathfrak{t}_k| \le 2|\delta_k| g_1(\tau_k)$ with $g_1 = \mathcal{M}(\varphi S'_{X_1})$, so $\langle \mathfrak{t}_k^4\rangle \le 16\langle \delta_k^8\rangle^{1/2}\langle g_1(\tau_k)^8\rangle^{1/2}$; here $\langle\delta_k^8\rangle \ll H_0^4/\log^8 T$ by (3.3) with $j=4$, and $\langle g_1(\tau_k)^8\rangle \ll (c_1\log T)^8$ by the sampling argument of Lemma 7.2 in $L^8$ (Sobolev on the intervals $\mathcal{U}_k$ with $g^8$, Theorem 3.10 with $p=8$, Lemma 8.2 with $r=2$). Hence $\langle \mathfrak{t}_k^4\rangle \ll (c_1^2 H_0)^2 = 1$, and $\chi_k \ll 1/T$. Dividing by $\mathcal{L}(\tau_k)^4 \asymp \log^4 T$,

$$\langle e_{1,k}^4\rangle_{\mathcal{I}_T} \ll \frac{H_1^2}{\log^4 T}. \tag{8.6}$$

*Transport of the second step.* The interval centred at $x_k^{(1)}$ of radius $|e_{1,k}|$ contains $\gamma_k$ and lies in $[T-2T^{1/2}, 2T+2T^{1/2}]$, so $|S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)})| \le 2|e_{1,k}| g_2(x_k^{(1)})$ with $g_2 = \mathcal{M}(\varphi S'_{X_2})$, and by Cauchy–Schwarz, (8.6) and Lemma 8.4 (applicable since $X_2 \le T^{1/8}$),

$$\Big\langle (S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)}))^2 \Big\rangle \le 4\langle e_{1,k}^4\rangle^{1/2}\langle g_2(x_k^{(1)})^4\rangle^{1/2} \ll \frac{H_1}{\log^2 T}(c_2\log T)^2 = c_2^2 H_1 = 1.$$

*Conclusion.* By (8.2) with $j=2$, $0 \le \chi_{2,k} \le e_{1,k}^2/(2\pi T)$, and $\langle \chi_{2,k}^2\rangle \ll T^{-2}$. Theorem 5.1 with $X = X_2$ (allowed, $2 \le X_2 \le T^{1/2}$) gives $\langle u_k(X_2)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_2} + O(1+\sqrt{\log(1/c_2)})$ with $\log\frac{1}{c_2} = \frac12 H_2$. Combining as in the proof of Theorem 7.3 gives the first statement of (8.5), and $\mathcal{L}(x_k^{(1)}) = \frac{\log T}{2\pi}(1+O(T^{-1/2}))$ on $\mathcal{I}_T$ gives the second. □

*8.5. All further steps: the exact identity, the local deformation, and the multiplicity*

This subsection proves the Newton steps $m \ge 3$ with constants uniform in $m$, up to $m = \log^* T + O(1)$ steps, where the error becomes $O(1)$ in units of zeros.

**Lemma 8.6** (The second-level identity)**.** *For $j \ge 1$, with $Y_k^{(j)} = \tau_k - x_k^{(j)}$ and $\chi''_{j,k} = \int_{x_k^{(j-1)}}^{\tau_k} (\mathcal{L}(\tau_k) - \mathcal{L}(y))\,dy$,*

$$\begin{aligned} Y_k^{(j)} &= \Big(1 - \frac{\mathcal{L}(\tau_k)}{\mathcal{L}(x_k^{(j-1)})}\Big) Y_k^{(j-1)} + \frac{S_{X_j}(x_k^{(j-1)}) + \chi''_{j,k}}{\mathcal{L}(x_k^{(j-1)})}, \\ \text{hence} \qquad \mathcal{L}(\tau_k) Y_k^{(j)} &= S_{X_j}(x_k^{(j-1)}) + R_{j,k}, \end{aligned} \tag{8.7}$$

*with $|R_{j,k}| \ll T^{c_j - 1}$ if $|Y_k^{(j-1)}| \le T^{c_{j-1}/2}$ and $c_{j-1} \le c_j$ (true by (2.11) and induction): the three contributions are $\mathcal{L}(\tau_k)(1-\mathcal{L}(\tau_k)/\mathcal{L}(x))Y_k^{(j-1)} \ll (Y_k^{(j-1)})^2/T$, $(\mathcal{L}(\tau_k)/\mathcal{L}(x)-1)S_{X_j}(x) \ll |Y_k^{(j-1)}|\sigma_{X_j}/(T\log T)$ and $\chi''_{j,k} \le (Y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$. Newton's step does not accumulate the corrections of the previous levels: it replaces the previous one by the phase of the new cutoff evaluated at the previous point.*

*Proof.* $N_0(x_k^{(j-1)}) - (k-\frac12) = N_0(x_k^{(j-1)}) - N_0(\tau_k) = -\mathcal{L}(\tau_k)Y_k^{(j-1)} + \chi''_{j,k}$, with $0 \le \chi''_{j,k} \le (Y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$; insert in (8.1) and use $1-\mathcal{L}(\tau_k)/\mathcal{L}(x) = O(|Y_k^{(j-1)}|/(T\log T))$ and $|S_{X_j}| \le \sigma_{X_j}$. □

**Proposition 8.7** (Local deformation of the second clock)**.** *Let $X_1 = T^{c_1}$, $X_2 = T^{c_2}$ with $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$, and $d_{j,k} = x_{k+1}^{(j)} - x_k^{(j)}$ ($d_{0,k} = \tau_{k+1} - \tau_k$). Then, for $T \ge T_0$,*

$$\Big\langle \Big| \frac{d_{2,k} - d_{0,k}}{d_{0,k}} \Big|^4 \Big\rangle_{\mathcal{I}_T} \ll c_2^4 = \frac{1}{H_1^2}, \qquad \frac{1}{\#\mathcal{I}_T} \#\Big\{ k \in \mathcal{I}_T : \Big| \frac{d_{2,k}}{d_{0,k}} - 1 \Big| > \alpha \Big\} \ll \frac{1}{\alpha^4 H_1^2} \quad (0 < \alpha \le 1), \tag{8.8}$$

*in particular $\#\{k \in \mathcal{I}_T : d_{2,k} \le 0\} \ll \#\mathcal{I}_T/(\log\log\log T)^2$: the second clock preserves the order and a spacing comparable to that of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices.*

*Proof.* By (8.7) with $j = 2$, $d_{2,k} - d_{0,k} = -\Delta_k\big(S_{X_2}(x^{(1)})/\mathcal{L}(\tau)\big) - \Delta_k(R_2/\mathcal{L}(\tau))$, where $\Delta_k a = a_{k+1} - a_k$; the last term is $\ll T^{c_1 - 1}$, negligible against $d_{0,k} \asymp 1/\log T$, and $\Delta_k(1/\mathcal{L}(\tau)) \ll 1/(T\log^3 T)$ is negligible too, so it suffices to bound $\Delta_k S_{X_2}(x^{(1)})/(d_{0,k}\mathcal{L}(\tau_k))$ in $\ell^4(\mathcal{I}_T)$. Let $g_j = \mathcal{M}(\varphi S'_{X_j})$ with $\varphi = 1$ on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$. Two applications of $|S(a) - S(b)| \le 2|a - b|\,\mathcal{M}(S')(a)$:

$$|x_{k+1}^{(1)} - x_k^{(1)}| \le d_{0,k} + |y_{k+1} - y_k| \le d_{0,k}\Big(1 + \frac{2g_1(\tau_k)}{\mathcal{L}(\tau_k)}\Big) + O\big(T^{c_1/2 - 1}\big),$$

$$|\Delta_k S_{X_2}(x^{(1)})| \le 2|x_{k+1}^{(1)} - x_k^{(1)}|\, g_2(x_k^{(1)}).$$

Hence, with $\mathcal{L}(\tau_k) \asymp \log T$,

$$\Big\| \frac{\Delta S_{X_2}(x^{(1)})}{d_0\,\mathcal{L}(\tau)} \Big\|_{4,\mathcal{I}_T} \ll \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{4,\mathcal{I}_T} + \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{8,\mathcal{I}_T} \Big\| \frac{g_1(\tau)}{\log T} \Big\|_{8,\mathcal{I}_T} + o(1).$$

Sampling at the $\tau_k$ in $L^8$ (Sobolev on the disjoint intervals $\mathcal{U}_k$ as in Lemma 7.2, with Lemma 8.2 for $\int |S'_{X_1}|^8$, $\int |S''_{X_1}|^8$ and Theorem 3.10 with $p = 8$) gives $\|g_1(\tau)/\log T\|_8 \ll c_1$. Sampling at the $x_k^{(1)} = x_k^{(1)}$ in $L^8$ is Lemma 8.4 with $F = g_2^8$: $\|F\|_2 = \|g_2\|_{16}^8$ and $\|F'\|_2 \le 8\|g_2\|_{16}^7 \|g_2'\|_{16}$, which need Theorem 3.7 with products of 8 primes ($X_2^{16} \le T$, true for $T \ge T_0$ since $c_2 \to 0$) and Theorem 3.10 with $p = 16$; the result is $\|g_2(x^{(1)})/\log T\|_8 \ll c_2$. The first inequality of (8.8) follows, and the second is Markov's inequality; the last statement is $\alpha = 1$. □

Proposition 8.7 controls consecutive differences; the multiplicity of the points $x_k^{(j)}$ at scale $\mathfrak{b}_T$, which Lemma 8.4 needs, is obtained for every $j$ by comparing $x_k^{(j)}$ with $\gamma_k$ through two unconditional counts of near pairs and near triples of ordinates. The ordinates enter the proof of the bound, as in Theorem 5.1, and not the construction of the points.

**Lemma 8.8** (Near pairs and triples of ordinates)**.** *Let $b = \mathfrak{b}_T$, $M = \#\mathcal{I}_T$ and, for $\mathsf{R} \ge 1$, $n_\Lambda(u) = \#\{l : |\gamma_l - u| \le \mathsf{R}b\}$. There are absolute constants $C_1, C_2$ such that for $T \ge T_0$ and every $\mathsf{R} \ge 1$,*

$$\#\{(k,l) \in \mathcal{I}_T^2 : |\gamma_k - \gamma_l| \le \mathsf{R}b\} \le C_1 \mathsf{R} M, \qquad \sum_{k \in \mathcal{I}_T} n_\Lambda(\gamma_k)^2 \le C_2 \mathsf{R}^2 M. \tag{8.9}$$

*Proof.* If $4\mathsf{R}b > 1$, then $N(u + h) - N(u) \le Ch\log T$ for $h \ge 1$ and $u \in [T - 1, 2T]$, by (2.3) and Theorem 3.1; hence every $\gamma_k$ has at most $C'\mathsf{R}$ ordinates within $\mathsf{R}b$, and both bounds follow. Let $4\mathsf{R}b \le 1$ and $h \in \{2\mathsf{R}b, 4\mathsf{R}b\}$. For $u \in [T - 1, 2T]$, $N(u + h) - N(u) =$

$h\mathcal{L}(u)+S(u+h)-S(u)+O(1/T)$ with $h\mathcal{L}(u)\le 5\mathsf{R}$, so Theorem 3.6 with $k=1,2$ and $\log(2+h\log T)\le\log(30\mathsf{R})\le 30\mathsf{R}$ give

$$\int_{T-1}^{2T}\Big(N(u+h)-N(u)\Big)^{2k}du\le C\mathsf{R}^{2k}T\qquad(k=1,2).$$

A pair $(k,l)$ with $|\gamma_k-\gamma_l|\le\mathsf{R}b$ lies in $[u,u+2\mathsf{R}b]$ for every $u\in[\min(\gamma_k,\gamma_l)-\mathsf{R}b,\min(\gamma_k,\gamma_l)]$, a set of measure $\mathsf{R}b$ contained in $[T-1,2T]$; hence the number of pairs is at most $\frac{1}{\mathsf{R}b}\int_{T-1}^{2T}(N(u+2\mathsf{R}b)-N(u))^2du\le C\mathsf{R}T/b\le C_1\mathsf{R}M$, as $M\asymp T/b$. A triple $(k,l,l')$ with $|\gamma_l-\gamma_k|,|\gamma_{l'}-\gamma_k|\le\mathsf{R}b$ lies in $[u,u+4\mathsf{R}b]$ for $u$ in a set of measure $2\mathsf{R}b$; by Hölder's inequality the number of triples is at most $\frac{1}{2\mathsf{R}b}\int_{T-1}^{2T}(N(u+4\mathsf{R}b)-N(u))^3du\le\frac{T^{1/4}(C\mathsf{R}^4T)^{3/4}}{2\mathsf{R}b}\le C_2\mathsf{R}^2M$. □

**Theorem 8.9** (Multiplicity of the corrected clock points)**.** *Let $j\ge0$ and let $x_k^{(j)}$ be given by* (8.1) *with cutoffs $X_i=T^{c_i}$, $c_i\le\frac12$, so that $x_k^{(j)}\in[T-2T^{1/2},2T+2T^{1/2}]$ for $k\in\mathcal{I}_T$ and $T\ge T_0$ (Lemma 8.6). Let $\tilde n_j(u)=\#\{k\in\mathcal{I}_T:u-\mathfrak{b}_T\le x_k^{(j)}\le u\}$, and let $\mathsf{A}\ge1$, $\mathsf{H}\ge1$ be such that $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T}\le\mathsf{A}\sqrt{\mathsf{H}}$. Then, with an absolute constant $C_3$,*

$$\int_{\mathbb{R}}\tilde n_j(u)^2\,du\le C_3\,\mathsf{A}\sqrt{\mathsf{H}}\;\mathfrak{b}_T\,\#\mathcal{I}_T.\tag{8.10}$$

*Proof.* Write $b=\mathfrak{b}_T$, $M=\#\mathcal{I}_T$, $x_k=x_k^{(j)}$, $e_k=e_{j,k}$. As in Lemma 8.3, $\int\tilde n_j^2=\sum_{k,l\in\mathcal{I}_T}(b-|x_k-x_l|)_+\le bP$ with $P=\#\{(k,l)\in\mathcal{I}_T^2:|x_k-x_l|<b\}$. Let $\mathsf{R}_0\ge2$. The pairs with $|\gamma_k-\gamma_l|\le\mathsf{R}_0b$ number at most $C_1\mathsf{R}_0M$ by (8.9). For the others, let $\mathsf{R}=2^i\mathsf{R}_0$, $i\ge0$, and consider the pairs with $\mathsf{R}b<|\gamma_k-\gamma_l|\le2\mathsf{R}b$ and $|x_k-x_l|<b$. For them $|e_k-e_l|\ge(\mathsf{R}-1)b\ge\mathsf{R}b/2$, and $b\mathcal{L}(\tau_k)\ge\frac12$ for $T\ge T_0$, so $\mathcal{L}(\tau_k)|e_k-e_l|\ge\mathsf{R}/4$ and

$$1\le\Big(\frac{4\mathcal{L}(\tau_k)|e_k-e_l|}{\mathsf{R}}\Big)^2\le\frac{32}{\mathsf{R}^2}\Big(\mathcal{L}(\tau_k)^2e_k^2+\mathcal{L}(\tau_l)^2e_l^2\Big)\Big(1+O(1/\log T)\Big).$$

Summing over the ordered pairs of the block and symmetrizing, their number $P_{\mathsf{R}}$ satisfies

$$P_{\mathsf{R}}\le\frac{65}{\mathsf{R}^2}\sum_{k\in\mathcal{I}_T}\mathcal{L}(\tau_k)^2e_k^2\,n_{2\mathsf{R}}(\gamma_k)\le\frac{65}{\mathsf{R}^2}\Big(\sum_k\mathcal{L}(\tau_k)^4e_k^4\Big)^{1/2}\Big(\sum_k n_{2\mathsf{R}}(\gamma_k)^2\Big)^{1/2}\le\frac{130\sqrt{C_2}\,\mathsf{A}^2\mathsf{H}}{\mathsf{R}}\,M,$$

by Cauchy–Schwarz, the hypothesis and (8.9). Summing over $i\ge0$ gives at most $260\sqrt{C_2}\,\mathsf{A}^2HM/\mathsf{R}_0$. With $\mathsf{R}_0=\mathsf{A}\sqrt{\mathsf{H}}\,(260\sqrt{C_2}/C_1)^{1/2}$ (which is $\ge2$ for $C_2$ large enough) the two parts are equal and $P\le2\sqrt{260C_1\sqrt{C_2}}\,\mathsf{A}\sqrt{\mathsf{H}}\,M$. □

Only the fourth moment of the error of the $j$-th clock enters. For $j=2$, with $\mathsf{A}$ bounded (shown below), the multiplicity of the second clock is $\ll\sqrt{H_2}\,\mathfrak{b}_T\#\mathcal{I}_T$. The factor $\sqrt{\mathsf{H}}$ is not a loss in the iteration: it is absorbed by the cutoff of the next step.

*All further steps*

Let $H_0=\log\log T$, $c_1=H_0^{-1/2}$, $c_2=H_1^{-1/2}$ with $H_1=\log H_0$, and define recursively, with an absolute constant $0<C_\star\le1$ fixed in the proof of Theorem 8.10,

$$H_j=2\log\frac{1}{c_j}\quad(j\ge1),\qquad c_j=C_\star\,H_{j-1}^{-9/16}\quad(j\ge3),\tag{8.11}$$

so that $H_2 = \log H_1$ as in Theorem 8.5, $H_j = \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$ for $j \ge 3$, and $\log H_{j-1} \le H_j$ for all $j$. All cutoffs depend only on $T$. Put

$$A_j(r) = \frac{\|\mathcal{L}(\tau_k)e_{j,k}\|_{2r,\mathcal{I}_T}}{\sqrt{H_j}} \quad (j \ge 1), \qquad A_0(r) = \frac{\|\mathcal{L}(\tau_k)\delta_k\|_{2r,\mathcal{I}_T}}{\sqrt{H_0}},$$

and let $m^* = m^*(T)$ be the first index $m$ with $H_m < H^*$, where $H^*$ is the absolute constant of the next theorem. Since $\log H_{j-1} \le H_j \le \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$, the sequence $H_j$ decreases from $\log\log T$ to below $H^*$ in $m^* = \log^* T + O(1)$ steps, where $\log^*$ is the iterated logarithm.

**Theorem 8.10** (All Newton steps, uniformly)**.** *There are absolute constants $C_\star$, $H^*$, $C$ and $T_0$ such that, with the cutoffs* (8.11)*, for every $T \ge T_0$ and every $1 \le m \le m^*$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{m,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_m}{4\pi^2} + \vartheta_m\, C\Big(1 + \sqrt{H_m}\Big), \qquad |\vartheta_m| \le 1, \tag{8.12}$$

*and $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T} \le C\sqrt{H_j}$ for $1 \le j < m^*$. In particular, at $m = m^* = \log^* T + O(1)$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(m^*)} - \gamma_k\Big)^2 \le \frac{C'}{\log^2 T}: \tag{8.13}$$

*the clock corrected $\log^* T + O(1)$ times with the Euler phases of the primes up to $T^{c_j}$ locates the ordinates with a mean-square error of $O(1)$ mean spacings, with an absolute constant, using no ordinate in its construction.*

*Proof. Multiplicities.* Let $\mathfrak{k}_j$ be the least $\kappa \ge 1$ with $\int \tilde{n}_j^2 \le \kappa \mathfrak{b}_T \#\mathcal{I}_T$. Then $\mathfrak{k}_0 = 1$ (the $\tau_k$ are separated by at least $\mathfrak{b}_T$, so only the diagonal contributes), $\mathfrak{k}_1 \le C$ (Lemma 8.3) and $\mathfrak{k}_j \le C_3 A_j(2)\sqrt{H_j}$ for $j \ge 2$ (Theorem 8.9 with $\mathsf{A} = \max(A_j(2), 1)$ and $\mathsf{H} = H_j$).

*Base.* By Lemma 8.2 with $X = 2$, $|S_2| \le \sigma_2 < 1$ and $0 \le \chi_k \le 1/T$, $\|\mathcal{L}(\tau_k)\delta_k\|_{2r} \le \|u_k(2)\|_{2r} + 2 \le C_0(r^2 + \sqrt{rH_0})$, so $A_0(r) \le C_0 r^2$ for $r \ge 1$.

*One step.* Let $j \ge 1$ and $r \ge 1$ with $X_j \le T^{1/(160r)}$. By (8.3), $\mathcal{L}(x_k^{(j-1)}) = \mathcal{L}(\tau_k)(1 + O(T^{-1/2}))$, Lemma 8.2 (with $V \le \log\frac{1}{c_j} + O(1) \le H_j$), Lemma 8.4 at the points $x_k^{(j-1)}$ with multiplicity constant $\mathfrak{k}_{j-1}$, and $\|\chi_j\|_{2r} \ll T^{c_{j-1}-1}$,

$$\sqrt{H_j}\,A_j(r) \le C\Big(r^2 + \sqrt{rH_j}\Big) + 2\sqrt{H_{j-1}}\,A_{j-1}(2r)\cdot\frac{C\mathfrak{k}_{j-1}^{1/(8r)}\sqrt{r}\,(1+\log X_j)}{\mathcal{L}(\tau_k)}\Big(1 + o(1)\Big) + o(1),$$

and $(1 + \log X_j)/\mathcal{L}(\tau_k) \le 2\pi c_j(1 + o(1))$, $\mathfrak{k}_{j-1}^{1/(8r)} \le \mathfrak{k}_{j-1}^{1/8}$. For $j \le 2$, $\sqrt{H_{j-1}}c_j = 1$ and $\mathfrak{k}_{j-1} \le C$; for $j \ge 3$, $\mathfrak{k}_{j-1}^{1/8}\sqrt{H_{j-1}}c_j \le (C_3 A_{j-1}(2))^{1/8} H_{j-1}^{1/16+1/2-9/16} C_\star = C_\star(C_3 A_{j-1}(2))^{1/8}$, and $A_{j-1}(2) \le A_{j-1}(2r)$ (monotonicity of the discrete norms). Fix $C_\star = \min(1, (4\pi C\, C_3^{1/8})^{-1})$. Then, for all $j \ge 1$ and $r \ge 1$,

$$A_j(r) \le C\Big(\sqrt{r} + \frac{r^2}{\sqrt{H_j}}\Big) + \frac{C_5\sqrt{r}}{\sqrt{H_j}}\max\Big(A_{j-1}(2r), A_{j-1}(2r)^{9/8}\Big), \tag{8.14}$$

with absolute $C, C_5$, provided $X_j \le T^{1/(160r)}$.

*Induction.* Let $B_\star = \max(2C, C_0, 1)$ and $C_6 = \big(4C/B_\star + 4C_5(4B_\star)^{9/8}/B_\star\big)^2$. We claim that for every $m \le m^*$,

$$A_j(r) \le B_\star r^2 \qquad (0 \le j \le m-1,\ 1 \le r \le 2^{m-j}). \tag{8.15}$$

For $j = 0$ this is the base. Suppose it holds at $j-1$ for $r \le 2^{m-j+1}$, let $1 \le r \le 2^{m-j}$, and assume $H_j \ge C_6 4^{m-j}$ and $X_j \le T^{1/(160r)}$. Since $A_{j-1}(2r) \le 4B_\star r^2$ and $4B_\star r^2 \ge 1$, (8.14) gives

$$A_j(r) \le C\sqrt{r} + \frac{Cr^2 + C_5(4B_\star)^{9/8} r^{11/4}}{\sqrt{H_j}} \le \frac{B_\star}{2} r^2 + \frac{\big(C + C_5(4B_\star)^{9/8}\big) r^3}{\sqrt{H_j}} \le B_\star r^2,$$

because $r^3/\sqrt{H_j} \le r^2 \cdot 2^{m-j}/\sqrt{H_j} \le r^2/\sqrt{C_6}$ and $\sqrt{C_6} \ge 2(C + C_5(4B_\star)^{9/8})/B_\star$.

*The two conditions.* Let $H^*$ be an absolute constant with $H^* \ge \max(4C_6, 160^2 \cdot 4)$ and so large that $e^{\frac{8}{9}(H - 2\log(1/C_\star))} \ge 4H$ and $e^H \ge 4H$ for $H \ge H^*$. If $m \le m^*$ then $H_{m-1} \ge H^*$ (by definition of $m^*$), and from $H_{j-1} \ge e^{H_j}$ $(j \le 2)$ or $H_{j-1} = e^{\frac{8}{9}(H_j - 2\log(1/C_\star))}$ $(j \ge 3)$ we get $H_{j-1} \ge 4H_j$ whenever $H_j \ge H^*$; by downward induction, $H_{m-1-i} \ge 4^i H^*$ for $0 \le i \le m-1$. Hence $H_j \ge 4^{m-1-j} H^* \ge C_6 4^{m-j}$ for $1 \le j \le m-1$, and $H_{j-1} \ge 4^{m-j} H^* \ge 160^2\, 4^{m-j}$ for $1 \le j \le m$, which gives $c_j \le H_{j-1}^{-1/2} \le 2^{-(m-j)}/160$, i.e. $X_j \le T^{1/(160r)}$ for $r \le 2^{m-j}$, and $X_j \ge 2$ because $c_j \ge C_\star(\log\log T)^{-9/16}$. So (8.15) holds for every $m \le m^*$, which is the statement $\|\mathcal{L} e_j\|_{4,\mathcal{I}_T} \le 4B_\star\sqrt{H_j}$ for $j < m^*$.

*The last step.* For $m \le m^*$ and $r = 1$, (8.15) at $j = m-1$ gives $A_{m-1}(2) \le 4B_\star$, so the transport term of the $m$-th step is $\|\mathcal{L}(\tau_k)\mathfrak{t}_k^{(m)}\|_{2,\mathcal{I}_T} \le C_5(4B_\star)^{9/8}$, with $\mathfrak{t}_k^{(m)} = S_{X_m}(\gamma_k) - S_{X_m}(x_k^{(m-1)})$. Theorem 5.1 at $X = X_m$ $(2 \le X_m \le T^{1/160})$ gives $\langle u_k(X_m)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_m} + O(1 + \sqrt{H_m}) = \frac{H_m}{4\pi^2} + O(1 + \sqrt{H_m})$, the cross term is at most $2\|u(X_m)\|_2\, C_5(4B_\star)^{9/8} \ll 1 + \sqrt{H_m}$, and $\chi_{m,k} \ll T^{-1/2}$; (8.2) gives (8.12). At $m = m^*$, $H_{m^*} < H^*$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(1/\log T))$ give (8.13). □

**Corollary 8.11** (Third step)**.** *With $c_3 = C_\star H_2^{-9/16}$ and $H_3 = \frac{9}{8}\log H_2 + 2\log(1/C_\star)$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{3,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_3}{4\pi^2} + O\big(1 + \sqrt{H_3}\big), \qquad \frac{1}{\#\mathcal{I}_T}\sum_{k \in \mathcal{I}_T}\big(x_k^{(3)} - \gamma_k\big)^2 = \frac{H_3 + O(\sqrt{H_3})}{\log^2 T},$$

*and the multiplicity of the second clock satisfies $\int \tilde{n}_2^2 \ll \sqrt{H_2}\,\mathfrak{b}_T \#\mathcal{I}_T$.*

*Proof.* Theorem 8.10 with $m = 3$ (for $T \ge T_0$, $H_2 \ge H^*$), and Theorem 8.9 with $\mathsf{A} = A_2(2) \le 4B_\star$ and $\mathsf{H} = H_2$. □

The cutoff of the third step, $T^{c_3}$ with $c_3 \asymp H_2^{-9/16}$ instead of $H_2^{-1/2}$, is what the multiplicity costs: the exponent $\frac{9}{16} = \frac{1}{2} + \frac{1}{16}$ comes from the factor $\sqrt{H_2}$ of (8.10) through the exponent $\frac{1}{8r} \le \frac{1}{8}$ of Lemma 8.4. The constants of Fujii's and Montgomery–Vaughan's bounds, which grow with the moment order $2^{m-j}$ used at level $j$, are absorbed by the normalization $H_j^{-1/2}$, because $H_j$ grows as a tower in $m - j$ while the orders grow geometrically.

A bounded error in units of zeros with a *fixed* final cutoff $T^{c_0}$ is not obtained by this method: the Newton term has mean square of order $c^2 \log\log T$ unless transport and defect cancel (Remark 7.7), and no estimate here proves such a cancellation.

## 9. Theorem E: the pointwise envelope

Let $\mathfrak{a} = \limsup_{t\to\infty} |S(t)|/\log t$. By Theorem 3.2, $\mathfrak{a} \le 0.10076$, and $\mathfrak{a} \le 0.1013$ also follows from [28]; whether $\mathfrak{a} = 0$ is not known.

**Proposition 9.1** (The envelope of the clock is the envelope of $S$)**.** *(a)* $\limsup_{k\to\infty} |\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *(b) If* $X_k \ge 2$ *and* $\sigma_{X_k} = o(\log\tau_k)$*, in particular if* $X_k \le (\log\tau_k)^{2-\varepsilon}$*, then* $\limsup_{k\to\infty} |\tau_k^{X_k} - \gamma_k| = 2\pi\mathfrak{a}$.

*Proof.* (a) The values $S(\gamma_k^{\pm})$ are limits of values of $S$ at non-ordinates. For $\varepsilon > 0$, (2.8) gives $|s_k| \le (\mathfrak{a}+\varepsilon)\log\gamma_k + O_\varepsilon(1)$, and $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log\gamma_k}{2\pi}(1+o(1))$ since $\delta_k = O(1)$; so $\limsup|\delta_k| \le 2\pi\mathfrak{a}$. Conversely, let $t > \gamma_1$ be a non-ordinate, let $\gamma < t < \gamma'$ be the nearest ordinates and $k = N(t)$, so that $\gamma_k = \gamma$ and $\gamma_{k+1} = \gamma'$. On $(\gamma,\gamma')$ the function $G$ is continuous and decreasing ($G' = -\mathcal{L}$), so $G(\gamma'^-) \le G(t) \le G(\gamma^+)$, while $s_k = k - \frac12 - N_0(\gamma) = G(\gamma^+) - \frac12$ and $s_{k+1} = k + \frac12 - N_0(\gamma') = G(\gamma'^-) + \frac12$. Hence $s_k \ge G(t) - \frac12$ and $s_{k+1} \le G(t) + \frac12$, so $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac12$, and $|G(t) - S(t)| = |\varpi(t)|/\pi \to 0$. Both ordinates are within bounded distance of $t$ (Theorem 3.1). Choosing non-ordinates $t \to \infty$ with $|S(t)| \ge (\mathfrak{a}-\varepsilon)\log t$ gives indices $k \to \infty$ with $|\delta_k| = |s_k|/\mathcal{L}(\xi_k) \ge 2\pi(\mathfrak{a}-\varepsilon) + o(1)$.

(b) $|\tau_k^{X_k} - \tau_k| = |S_{X_k}(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_{X_k}/\mathcal{L}(\tau_k) \to 0$, and $\sigma_X \ll \sqrt{X}/\log X$. □

So the corrected clocks with slowly growing cutoffs have exactly the pointwise envelope of the bare clock, and for every cutoff the envelope of $\tau_k$ is that of $S$. A smaller coefficient for the clock is a smaller coefficient for $S(t)$.

**Theorem 9.2** (Theorem E)**.** *Let* $\mathfrak{e}$ *be continuous on* $[\frac12, 1]$ *with* $\mathfrak{e}(1) = 0$*, such that for every* $\varepsilon > 0$, $\zeta(\sigma+it) \ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon}$ *uniformly for* $\frac12 \le \sigma \le 1$ *and* $|t| \ge 2$*, and put* $\mathfrak{e}(\sigma) = \frac12 - \sigma + \mathfrak{e}(1-\sigma)$ *for* $0 \le \sigma < \frac12$*. Then*

$$\mathfrak{a} \le \frac{1}{2\pi\log 2}\int_0^1 \frac{\mathfrak{e}(\sigma)}{\sqrt{\sigma(2-\sigma)}}\,d\sigma. \tag{9.1}$$

*For* $J \ge 0$ *let* $(\mathsf{k}_0, \mathsf{l}_0) = (\frac{13}{84}, \frac{55}{84})$, $(\mathsf{k}_j, \mathsf{l}_j) = A^j(\mathsf{k}_0, \mathsf{l}_0)$ *with* $A(\mathsf{k},\mathsf{l}) = \left(\frac{\mathsf{k}}{2\mathsf{k}+2}, \frac12 + \frac{\mathsf{l}}{2\mathsf{k}+2}\right)$*, and let* $\mathfrak{e}_J$ *be the piecewise linear function through the points* $(\mathsf{l}_j - \mathsf{k}_j, \mathsf{k}_j)$, $0 \le j \le J$*, and* $(1,0)$*; the first vertices are* $(\frac12, \frac{13}{84})$, $(\frac{139}{194}, \frac{13}{194})$, $(\frac{173}{207}, \frac{13}{414})$, $(\frac{773}{854}, \frac{13}{854})$*. Let* $C_J$ *be the right side of* (9.1) *for* $\mathfrak{e} = \mathfrak{e}_J$*. Then* $\mathfrak{a} \le C_J$ *for every* $J$*;* $C_J$ *decreases,* $C_7 = 0.0968160063\ldots$ *and* $C_{30} = 0.0968151142790\ldots$*; hence*

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| = 2\pi\mathfrak{a} \le 2\pi C_{30} = 0.6083073\ldots < 0.60831. \tag{9.2}$$

*Proof. Uniform bounds.* Extend $\mathfrak{e}$ by 0 on $[1,\infty)$ and by $\frac12 - \sigma$ on $(-\infty, 0)$; the extension $\mathfrak{e}^*$ is continuous ($\mathfrak{e}(0) = \frac12 + \mathfrak{e}(1) = \frac12$). For $\sigma \ge 1$, $\zeta(\sigma+it) \ll \log|t|$ uniformly [27, Thm. 3.5]. For $-\frac{1}{50} \le \sigma < \frac12$, $\zeta(\sigma+it) = \chi(\sigma+it)\zeta(1-\sigma-it)$ with $|\chi(\sigma+it)| = (|t|/2\pi)^{1/2-\sigma}(1+O(1/|t|))$ uniformly [27, (4.12.3)], and $|\zeta(1-\sigma-it)| = |\zeta(1-\sigma+it)|$. Hence $\zeta(\sigma+it) \ll_\varepsilon |t|^{\mathfrak{e}^*(\sigma)+\varepsilon}$ uniformly for $-\frac{1}{50} \le \sigma \le 4$, $|t| \ge 2$.

*Backlund's method.* Let $0 < \eta < \frac{1}{100}$, $\sigma_\circ = 1+\eta$, $\varrho = 1+2\eta$. Then $1 < \sigma_\circ < \varrho$, $\sigma^\sharp = \frac54 + \frac32\eta$, $\delta^\sharp = \frac14 + \frac{\eta}{2} \in (0, \frac12)$, and $\varrho/(\sigma_\circ - \frac12) = 2$. For $s = \sigma_\circ + \varrho e^{i\theta}$, $\Re s \in [-\eta, 2+3\eta]$ and $|\Im s| \le 2$, so for $T$ large

$$\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(s)| \le \log\max_{\pm}|(s\pm iT-1)\zeta(s\pm iT)| \le \log(T+4) + (\mathfrak{e}^*(\Re s)+\varepsilon)\log(T+2) + C_\varepsilon =: F_\circ(\theta),$$

an even continuous function. By Theorem 3.14(a), along $\varkappa_m \to \infty$,

$$\limsup_m \mathcal{Z}_{\varkappa_m}(\sigma_\circ, \varrho) \le \log T \cdot \frac{1}{\pi}\int_0^\pi \Big(\mathfrak{e}^*(\sigma_\circ + \varrho\cos\theta) + \varepsilon\Big)d\theta + O_{\varepsilon,\eta}(1),$$

the terms $\log(T+4)$ and $-\log\sqrt{\eta^2+T^2}$ cancelling up to $O(1/T)$. The argument defining $S(T)$ runs along $2 \to 2+iT \to \sigma^\sharp + iT \to \frac12 + iT$, and on the first two segments $\Re s > 1$, where $\log\zeta$ is holomorphic; so $\pi S(T) = \arg\zeta(\sigma^\sharp + iT) + \Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT)) - \Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1)$, with $|\arg\zeta(\sigma^\sharp+iT)| \le \log\zeta(\sigma^\sharp)$ and the last term bounded by $\pi$. Theorem 3.14(b) with $\varkappa = \varkappa_m \to \infty$ gives, for $T$ large,

$$|S(T)| \le \frac{\log T}{2\log 2}\cdot\frac{1}{\pi}\int_0^\pi \Big(\mathfrak{e}^*(1+\eta+(1+2\eta)\cos\theta) + \varepsilon\Big)d\theta + O_{\varepsilon,\eta}(1).$$

*Limit.* Divide by $\log T$, let $T\to\infty$, then $\varepsilon\to 0$ and $\eta\to 0$; $\mathfrak{e}^*$ is bounded and continuous, so by dominated convergence $\mathfrak{a} \le \frac{1}{2\pi\log 2}\int_0^\pi \mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac{\pi}{2}$; on $[\frac{\pi}{2},\pi]$ substitute $\sigma = 1+\cos\theta$, $d\sigma = -\sin\theta\,d\theta$, $\sin\theta = \sqrt{\sigma(2-\sigma)}$, which gives (9.1).

*The profile.* Let $\varepsilon' > 0$, $(\mathsf{k}_0,\mathsf{l}_0) = (\frac{13}{84}+\varepsilon', \frac{55}{84}+\varepsilon')$ and $(\mathsf{k}_j,\mathsf{l}_j) = A^j(\mathsf{k}_0,\mathsf{l}_0)$ for $1\le j\le J$, exponent pairs by Theorem 3.12(a),(c), with $\mathsf{l}_j - \mathsf{k}_j \ge \frac12$; by (b), $|\zeta(\mathsf{l}_j - \mathsf{k}_j + it)| \ll |t|^{\mathsf{k}_j}\log|t|$, and $|\zeta(1+it)| \ll \log|t|$. The points $(\mathsf{l}_j-\mathsf{k}_j, \mathsf{k}_j)$ and $(1,0)$ have decreasing ordinates, and Theorem 3.13 between consecutive points gives the hypothesis of the theorem for the piecewise linear function through them. For $\varepsilon' = 0$ these are the vertices of $\mathfrak{e}_J$; the right side of (9.1) depends continuously on $\varepsilon'$, so $\mathfrak{a} \le C_J$. On each linear piece $\mathfrak{e}(\sigma) = \alpha\sigma+\beta$ of $[0,1]$ a primitive of $\mathfrak{e}(\sigma)/\sqrt{\sigma(2-\sigma)}$ is $-\alpha\sqrt{\sigma(2-\sigma)} + (\alpha+\beta)\arcsin(\sigma-1)$, and the values of $C_J$ at the rational vertices are computed from it at 50 digits by `opc_envelope_iterations_p1.py`; the value $C_7$ agrees with the quadrature of `opc_envelope_coefficient_v14.py`. Each new vertex lies below the chord from the previous one to $(1,0)$, so $C_J$ decreases. Multiply by $2\pi$ and use Proposition 9.1(a). □

*Remark* 9.3. (a) Trudgian [28, §5] obtained $\mathfrak{a} \le 0.1013$ from Huxley's exponent $\frac{32}{205}$ at $\sigma = \frac12$ by the same method; Theorem 9.2 replaces the exponents and lets the parameters of the disc tend to their limits. With the explicit exponents used by Bellotti and Wong ($\frac{27}{164}$ at $\sigma=\frac12$, Yang's lines $\sigma_j = 1 - j/(2^j-2)$ with exponents $1/(2^j-2)$, $4\le j\le 9$ [1]) and linear interpolation between $\sigma=0$ and $\sigma=\frac12$, as in [1, §3.1], the right side of (9.1) is 0.10074, the limit of their $C_1 = 0.10076$ as their parameters tend to $(1,1)$; the same right-hand profile reflected by the functional equation gives 0.09783. (b) Profiles of $\mu(\sigma)$ below $\mathfrak{e}_J$ are known: Trudgian and Yang [30] close a convex hull of known exponent pairs, Bourgain's among them, under the $A$-, $B$- and $C$-processes and deduce piecewise bounds for $\mu(\sigma)$ on $[\frac12,1]$. Any such profile can be inserted in (9.1); we do not evaluate it here. (c) For the Lindelöf profile ($\mathfrak{e}=0$ on $[\frac12,1]$) the right side is 0.07863: Backlund's method with a disc cannot give less, whatever the exponents. (d) Theorem 9.2 is asymptotic: the explicit constants of the bounds for $\zeta$ do not enter, which is why non-explicit exponents may be used, and it does not replace the explicit bounds of [1], valid for all $T \ge e$.

## 10. Conclusions

### *10.1. What was known*

Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup|\delta_k| \le 0.6331$ from the explicit bound of Bellotti and Wong [1], and $\limsup|\delta_k| \le 0.6365$ from Trudgian's asymptotic bound [28]. On average, $s_k$ is asymptotically Gaussian with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [17, 16], so $\langle\delta_k^2\rangle \sim 2\log\log T/\log^2 T$: the clock is close to the zeros in height but misplaces almost every zero by an unbounded number of positions. Selberg's approximation [23, 11] replaces $S(t)$ by short prime sums in $t$-mean, and França–LeClair and LeClair [9, 18, 10] write the equation of an individual zero with an Euler phase and a growing number of primes, without an error estimate over the zeros.

### *10.2. What this paper adds*

1. *An exact dictionary between clock-type models and the zeros* (Theorems 4.1 and 4.2). For any $C^1$ model $F$ and any continuous $\mathfrak{g}$, sums of $\mathfrak{g}$-increments of $k - \frac{1}{2} - F(\gamma_k)$ over the ordinates are exactly $dF$-integrals of $\mathfrak{g}(N - F)$ up to two boundary terms; for powers this gives the even moments with the constant $\frac{1}{12}$ per zero, for exponentials it transfers the signed law of $S(t)$ to $s_k$ (Corollary 4.4).

2. *The prime information needed by the clock* (Theorems 5.1 and 5.12). The corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log(\log T/\log X)})$, also with a cutoff varying at every index; the defect is bounded in mean square if and only if the primes reach a fixed power of the height.

3. *An exact finite-height account* (Theorem 6.1). The gain of the one-step clock is a diagonal prime response minus a cross term, and its residual differs from the defect by a transport term and an explicit nonnegative second-order term.

4. *A computable clock that approaches the zeros* (Theorems 7.3, 8.5 and 8.10). With $X = T^{1/\sqrt{\log\log T}}$ the one-step clock has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in height, against $(2 + o(1))\log\log T/\log^2 T$ for the bare clock; the $m$-th Newton step has error $\frac{H_m}{4\pi^2} + O(1 + \sqrt{H_m})$ in units of zeros with absolute constants, and after $\log^* T + O(1)$ steps the error is $O(1)$ mean spacings, using no ordinate. The proof needs an exact moment recursion, a multiplicity bound for the corrected points of every level (Theorem 8.9) and a cutoff law that absorbs the multiplicity.

5. *The pointwise envelope* (Proposition 9.1, Theorem 9.2). $\limsup|\tau_k - \gamma_k| = 2\pi\limsup|S(t)|/\log t$ exactly, no slowly growing prime correction changes it, and Backlund's method with the profile generated from Bourgain's exponent pair gives $\limsup|\tau_k - \gamma_k| < 0.60831$.

### *10.3. Open problems*

**Question 10.1.** For fixed $c \in (0, \frac{1}{2}]$, do the bounded means of $u_k(T^c)^2$ converge, and to $(2\pi^2)^{-1}\log(1/c)$ plus which constant?

**Question 10.2.** Does $\langle r_k(T^c)^2\rangle_{\mathcal{I}_T}$ stay bounded for fixed $c$, that is, does the transport cancel against the defect? Is the error of the corrected clock after $\log^* T + O(1)$ steps bounded in units of zeros for *almost every* index, and not only in mean square? Can Theorems 5.1 and 7.3 be extended to $c > \frac{1}{2}$?

**Question 10.3.** Can the absolute constants $C_\star$, $H^*$ and $B_\star$ of Theorem 8.10, and those of Fujii's and Montgomery–Vaughan's bounds on which they rest, be made explicit, so that the cutoff law (8.11) becomes a formula that can be evaluated at a given height?

Theorem 4.1 applies to any counting staircase and any $C^1$ model, in particular to the zeros of a primitive Dirichlet $L$-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s,\chi)$ [24, 11] and uniformity in the conductor.

## Declaration of competing interest

The author declares no competing interests.

## Data availability

No data were used: every statement is proved. The two programs of Appendix Appendix A evaluate the integral (9.1) for piecewise linear exponent profiles with rational vertices; they and their outputs are in the public GitHub repository `https://github.com/vcasana1/LawCasanya`, directory `work/paper1/`. They used Python 3 and mpmath 1.3.0.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI assistant (Claude, Anthropic) to edit the text, check proofs and write programs. The author reviewed and edited the content and takes full responsibility for it.

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## Appendix A. Programs

The two programs below are in the directory `work/paper1/code/` of the repository and are reproduced verbatim. They read no data. The first, `opc_envelope_coefficient_v14.py`, evaluates (9.1) by 40-digit quadrature for the profiles of Remark 9.3 and for $\mathfrak{e}_7$; the second, `opc_envelope_iterations_p1.py`, evaluates it in closed form for $\mathfrak{e}_J$, $0 \le J \le 30$. They run under Python 3 with mpmath.

### *Appendix A.1.* `opc_envelope_coefficient_v14.py`

```
#!/usr/bin/env python3
"""Coefficient of the asymptotic envelope limsup |S(t)|/log t (Theorem E of ordinal_prime_cutoff_v14).

Usage:  python3 opc_envelope_coefficient_v14.py OUT_DIR

For an exponent profile e(sigma) on [0,1] (zeta(sigma+it) << |t|^{e(sigma)+eps} uniformly), the
Backlund-Jensen bound of Bellotti-Wong (Lemmas 2.1 and 2.3 of arXiv:2412.15470), with the disc centred
at b and radius rho tending to (1,1), gives

    limsup |S(t)|/log t <= C(e) = 1/(2 pi log 2) * int_0^1 e(sigma) / sqrt(sigma (2 - sigma)) d sigma .

The script evaluates C(e) for four profiles, with 40-digit quadrature on each linear piece:
  (a) the profile of Bellotti-Wong: 27/164 at sigma=1/2, Yang's lines sigma_k=1-k/(2^k-2) (k=4..9) with
      exponent 1/(2^k-2), 0 at sigma=1, linear interpolation between 0 and 1/2 (no reflection);
  (b) the same right-hand profile, reflected to [0,1/2] by the functional equation;
  (c) Bourgain's exponent pair (13/84,55/84) and its images A^j (j=1..7) under van der Corput's
      A-process, lower convex hull, reflected to [0,1/2];
  (d) the Lindelof profile (floor of the method).
It also evaluates the general disc (b, rho) for profile (a) as a check against 0.10076.
No data are read.
"""
import json
import sys
from fractions import Fraction as Fr

import mpmath as mp

mp.mp.dps = 40


def lower_hull(points):
    pts = sorted(set(points))
    hull = []
    for p in pts:
        while len(hull) >= 2:
            (x1, y1), (x2, y2) = hull[-2], hull[-1]
            if (y2 - y1) * (p[0] - x1) >= (p[1] - y1) * (x2 - x1):
                hull.pop()
            else:
                break
        hull.append(p)
    return hull


def piecewise(points):
    xs = [mp.mpf(x.numerator) / x.denominator if isinstance(x, Fr) else mp.mpf(x) for x, _ in points]
    ys = [mp.mpf(y.numerator) / y.denominator if isinstance(y, Fr) else mp.mpf(y) for _, y in points]

    def f(s):
        for i in range(len(xs) - 1):
            if xs[i] <= s <= xs[i + 1]:
                return ys[i] + (ys[i + 1] - ys[i]) * (s - xs[i]) / (xs[i + 1] - xs[i])
        raise ValueError(s)
    return f, xs

```

```
56
57 def profile(right_points, reflect, left_value_at_half=None):
58     """Return e(sigma) on [0,1] and the list of breakpoints."""
59     right, xs = piecewise(right_points)
60     half = mp.mpf(1) / 2
61
62     def e(s):
63         if s >= 1:
64             return mp.mpf(0)
65         if s >= half:
66             return right(s)
67         if reflect:
68             return half - s + right(1 - s)
69         # linear interpolation between e(0)=1/2 and e(1/2)
70         return (1 - 2 * s) * half + 2 * s * left_value_at_half
71     bps = sorted(set([mp.mpf(0), half, mp.mpf(1)] + xs + ([1 - x for x in xs] if reflect else [])))
72     return e, bps
73
74
75 def coefficient(e, bps):
76     w = lambda s: e(s) / mp.sqrt(s * (2 - s))
77     total = mp.mpf(0)
78     for a, b in zip(bps[:-1], bps[1:]):
79         if b > a:
80             total += mp.quad(w, [a, b])
81     return total / (2 * mp.pi * mp.log(2))
82
83
84 def disc_coefficient(e, b, rho):
85     """(1/pi) int_0^pi e(b + rho cos th) dth / (2 log(rho/(b-1/2))), e = 0 for sigma >= 1."""
86     g = lambda th: e(min(max(b + rho * mp.cos(th), mp.mpf(0)), mp.mpf(1))) if b + rho * mp.cos(th) >= 0
       ↪ else mp.mpf(1) / 2 - (b + rho * mp.cos(th))
87     val = mp.quad(g, mp.linspace(0, mp.pi, 64)) / mp.pi
88     return val / (2 * mp.log(rho / (b - mp.mpf(1) / 2)))
89
90
91 def main(out_dir):
92     half = Fr(1, 2)
93     yang = [(Fr(1) - Fr(k, 2 ** k - 2), Fr(1, 2 ** k - 2)) for k in range(4, 10)]
94     bw_right = lower_hull([(half, Fr(27, 164))] + yang + [(Fr(1), Fr(0))])
95     e_a, b_a = profile(bw_right, reflect=False, left_value_at_half=mp.mpf(27) / 164)
96     e_b, b_b = profile(bw_right, reflect=True)
97
98     pairs = []
99     k, l = Fr(13, 84), Fr(55, 84)
100    pairs.append((k, l))
101    for _ in range(7):
102        k, l = k / (2 * k + 2), half + l / (2 * k + 2)
103        pairs.append((k, l))
104    bourgain_points = [(l - k, k) for k, l in pairs] + [(Fr(1), Fr(0))]
105    bourgain_right = lower_hull(bourgain_points)
106    e_c, b_c = profile(bourgain_right, reflect=True)
107
108    lindelof_right = [(half, Fr(0)), (Fr(1), Fr(0))]
109    e_d, b_d = profile(lindelof_right, reflect=True)
110
111    res = {
112        "formula": "C(e) = (1/(2 pi log 2)) * int_0^1 e(sigma)/sqrt(sigma(2-sigma)) dsigma",
113        "a_BellottiWong_profile_no_reflection": float(coefficient(e_a, b_a)),
114        "b_same_profile_with_reflection": float(coefficient(e_b, b_b)),
115        "c_Bourgain_Aimages_with_reflection": float(coefficient(e_c, b_c)),
116        "d_Lindelof_floor": float(coefficient(e_d, b_d)),
117        "check_disc_b1.0001_rho1.0002_profile_a": float(disc_coefficient(e_a, mp.mpf("1.0001"), mp.mpf("
       ↪ 1.0002"))),
118        "Bourgain_hull_points": [[str(x), str(y), float(x), float(y)] for x, y in bourgain_right],
119        "BW_hull_points": [[str(x), str(y), float(x), float(y)] for x, y in bw_right],
120        "exponent_pairs_used": [[str(k), str(l)] for k, l in pairs],
121    }
```

```
    for key in ("a_BellottiWong_profile_no_reflection", "b_same_profile_with_reflection",
                "c_Bourgain_Aimages_with_reflection", "d_Lindelof_floor"):
        res[key + "_times_2pi"] = float(2 * mp.pi * res[key])
    res["c_exact_40_digits"] = mp.nstr(coefficient(e_c, b_c), 30)
    res["c_times_2pi_40_digits"] = mp.nstr(2 * mp.pi * coefficient(e_c, b_c), 30)
    with open(f"{out_dir}/envelope_coefficient.json", "w") as fh:
        json.dump(res, fh, indent=1)
    for key, val in res.items():
        if not isinstance(val, list):
            print(f"{key}: {val}")


if __name__ == "__main__":
    main(sys.argv[1] if len(sys.argv) > 1 else "outputs")
```

## *Appendix A.2.* `opc_envelope_iterations_p1.py`

```
#!/usr/bin/env python3
"""Envelope coefficient C_J of Theorem E for the profile built from Bourgain's exponent pair (13/84, 55/84)
by J applications of van der Corput's A-process, A(k,l) = (k/(2k+2), 1/2 + l/(2k+2)), the points
(l_j - k_j, k_j), j = 0..J, joined linearly to (1,0) and reflected by the functional equation:
   C(e) = (1/(2 pi log 2)) int_0^1 e(sigma) / sqrt(sigma (2 - sigma)) d sigma,
   e(sigma) = 1/2 - sigma + e(1 - sigma) on [0, 1/2).
Exact rational vertices; each linear piece integrated in closed form with mpmath at 50 digits:
   int (alpha s + beta) / sqrt(s(2-s)) ds = -alpha sqrt(s(2-s)) + (alpha + beta) arcsin(s - 1).
For every J the bound a <= C_J holds (Theorem E); C_J decreases with J.  Also checks that each new vertex lies
below the chord from the previous vertex to (1,0).  No data are read.
Usage: python3 opc_envelope_iterations_p1.py [OUT_DIR]   (writes envelope_iterations_p1.{json,txt})
"""
import json, os, sys
from fractions import Fraction as Fr
import mpmath as mp

mp.mp.dps = 50
OUT = sys.argv[1] if len(sys.argv) > 1 else os.path.join(os.path.dirname(os.path.abspath(__file__)), '../outputs')


def vertices(J):
    k, l = Fr(13, 84), Fr(55, 84)
    P = [(1 - k, k)]
    for _ in range(J):
        k, l = k / (2 * k + 2), Fr(1, 2) + l / (2 * k + 2)
        P.append((1 - k, k))
    P.append((Fr(1), Fr(0)))
    return P


def prim(alpha, beta, s):
    s = mp.mpf(s)
    return -alpha * mp.sqrt(s * (2 - s)) + (alpha + beta) * mp.asin(s - 1)


def piece_integral(a, ea, b, eb):
    alpha = mp.mpf(eb - ea) / mp.mpf(b - a)
    beta = mp.mpf(ea) - alpha * mp.mpf(a)
    return prim(alpha, beta, b) - prim(alpha, beta, a)


def coefficient(P):
    total = mp.mpf(0)
    for (a, ea), (b, eb) in zip(P, P[1:]):          # right half [1/2, 1]
        total += piece_integral(a, ea, b, eb)
    # left half: e(s) = 1/2 - s + e(1-s); vertices reflected: s' = 1 - a, value 1/2 - s' + ea (linear in s')
    for (a, ea), (b, eb) in zip(P, P[1:]):
```

```
        sa, sb = 1 - b, 1 - a                       # interval [1-b, 1-a] in [0, 1/2]
        va, vb = Fr(1, 2) - sa + eb, Fr(1, 2) - sb + ea
        total += piece_integral(sa, va, sb, vb)
    return total / (2 * mp.pi * mp.log(2))


rows, lines = [], []
prev = None
for J in range(0, 31):
    P = vertices(J)
    below = None
    if J >= 1:
        (s0, k0), (s1, k1) = P[-3], P[-2]
        below = bool(k1 < k0 * (1 - s1) / (1 - s0))
    c = coefficient(P)
    rows.append(dict(J=J, last_vertex=[str(P[-2][0]), str(P[-2][1])], below_chord=below,
                     C_J=mp.nstr(c, 20), two_pi_C_J=mp.nstr(2 * mp.pi * c, 20),
                     decrease=None if prev is None else mp.nstr(prev - c, 6)))
    lines.append(f"J={J:2d}  sigma_J={float(P[-2][0]):.8f}  below_chord={below}  C_J={mp.nstr(c, 16)}  "
                 f"2pi*C_J={mp.nstr(2 * mp.pi * c, 12)}" + ("" if prev is None else f"  decrease={mp.nstr(
     ↪ prev - c, 4)}"))
    prev = c
os.makedirs(OUT, exist_ok=True)
json.dump(rows, open(os.path.join(OUT, 'envelope_iterations_p1.json'), 'w'), indent=1)
open(os.path.join(OUT, 'envelope_iterations_p1.txt'), 'w').write('\n'.join(lines) + '\n')
print('\n'.join(lines))
```